Appendix A
Solutions to Problems

1.1. Since $\widetilde{E}_{\rm rel}(\widehat{x}) = E_{\rm rel}(\widehat{x})|x|/|\widehat{x}|$ and $\widehat{x} = x(1 \pm E_{\rm rel}(\widehat{x}))$, we have

$$\frac{E_{\rm rel}(\widehat{x})}{1 + E_{\rm rel}(\widehat{x})} \le \widetilde{E}_{\rm rel}(\widehat{x}) \le \frac{E_{\rm rel}(\widehat{x})}{1 - E_{\rm rel}(\widehat{x})}.$$

Hence if $E_{\rm rel}(\widehat{x}) < 0.01$, say, then there is no difference between $E_{\rm rel}$ and $\widetilde{E}_{\rm rel}$ for practical purposes.

1.2. Nothing can be concluded about the last digit before the decimal point. Evaluating y to higher precision yields

t	y
35	262537412640768743.99999999999925007
40	262537412640768743.9999999999992500725972

This shows that the required digit is, in fact, 3. The interesting fact that y is so close to an integer was pointed out by Lehmer [779, 1943], who explains its connection with number theory. It is known that y is irrational [1090, 1991].

1.3.
1. $x/(\sqrt{x+1}+1)$.
2. $2\sin\frac{x-y}{2}\cos\frac{x+y}{2}$.
3. $(x-y)(x+y)$. Cancellation has not been avoided, but it is now harmless if x and y are known exactly (see also Problem 3.8).
4. $\sin x/(1+\cos x)$.
5. $c = \big((a-b)^2 + ab(2\sin\theta/2)^2\big)^{1/2}$.

1.4. $a+ib = (x+iy)^2 = x^2-y^2+2ixy$, so $b = 2xy$ and $a = x^2-y^2$, giving $x^2 - b^2/(4x^2) = a$, or $4x^4 - 4ax^2 - b^2 = 0$. Hence

$$x^2 = \frac{4a \pm \sqrt{16a^2+16b^2}}{8} = \frac{a \pm \sqrt{a^2+b^2}}{2}.$$

If $a \ge 0$ we use this formula with the plus sign, since $x^2 \ge 0$. If $a < 0$ this formula is potentially unstable, so we use the rewritten form

$$x^2 = \frac{b^2}{2(-a+\sqrt{a^2+b^2})}.$$

In either case we get two values for x and recover y from $y = b/(2x)$. Note that there are other issues to consider here, such as scaling to avoid overflow.

1.5. A straightforward evaluation of $\log(1+x)$ is not sufficient, since the addition $1+x$ loses significant digits of x when x is small. The following method is effective (for another approach, see Hull, Fairgrieve, and Tang [649, 1994]): calculate $w = 1+x$ and then compute

$$f(x) = \begin{cases} x, & w = 1, \\ \dfrac{x \log w}{w-1}, & w \neq 1. \end{cases} \tag{A.1}$$

The explanation of why this method works is similar to that for the method in §1.14.1. We have $\widehat{w} = (1+x)(1+\delta)$, $|\delta| \le u$, and if $\widehat{w} = 1$ then $|x| \le u + u^2 + u^3 + \cdots$, so from the Taylor series $f(x) = x(1 - x/2 + x^2/3 - \cdots)$ we see that $f(x) = x$ is a correctly rounded result. If $\widehat{w} \neq 1$ then

$$\widehat{f} = \frac{(x \log \widehat{w})(1+\epsilon_1)(1+\epsilon_2)}{(\widehat{w}-1)(1+\epsilon_3)}(1+\epsilon_4), \qquad |\epsilon_i| \le u.$$

Defining $\widehat{w} =: 1+z$,

$$g(\widehat{w}) := \frac{\log \widehat{w}}{\widehat{w}-1} = \frac{\log(1+z)}{z} = 1 - z/2 + z^2/3 - \cdots,$$

so if $\widehat{w} \approx w \approx 1$ (thus $x \approx 0$) then

$$g(w) - g(\widehat{w}) \approx \frac{\widehat{w}-w}{2} = \frac{(1+x)\delta}{2} \approx \frac{\delta}{2} \approx \frac{g(w)\delta}{2}.$$

Thus, with $1+\theta := (1+\epsilon_1)(1+\epsilon_2)(1+\epsilon_3)^{-1}(1+\epsilon_4)$,

$$\widehat{f} = xg(\widehat{w})(1+\theta) \approx xg(w)(1+\theta) = f(x)(1+\theta),$$

showing that $\widehat{f}$ is an accurate approximation to $f(x) = \log(1+x)$. Figure A.1 shows MATLAB plots of $\log(1+x)$, using MATLAB's built-in `log` function, and `log1p`, an M-file of Cleve Moler that implements (A.1). Close to the origin the lack of smoothness of the curve for the `log` evaluation is clear.

1.7. From (1.4) we have

$$\begin{aligned} (n-1)V(x+\Delta x) &= \sum_{i=1}^{n}\Big(x_i + \Delta x_i - \overline{x} - n^{-1}\sum_{j=1}^{n}\Delta x_j\Big)^2 \\ &= (n-1)V(x) + 2\sum_{i=1}^{n}(x_i - \overline{x})\Big(\Delta x_i - n^{-1}\sum_{j=1}^{n}\Delta x_j\Big) \\ &\qquad + \sum_{i=1}^{n}\Big(\Delta x_i - n^{-1}\sum_{j=1}^{n}\Delta x_j\Big)^2 \\ &= (n-1)V(x) + 2\sum_{j=1}^{n}(x_i - \overline{x})\Delta x_i \\ &\qquad + \sum_{i=1}^{n}\Big(\Delta x_i - n^{-1}\sum_{j=1}^{n}\Delta x_j\Big)^2. \end{aligned}$$

This yields the following inequality, which is attainable to first order:

$$(n-1)|V(x) - V(x+\Delta x)| \le 2\epsilon\sum_{i=1}^{n}|x_i - \overline{x}||x_i| + \epsilon^2\sum_{i=1}^{n}\Big(|x_i| + n^{-1}\sum_{j=1}^{n}|x_j|\Big)^2.$$

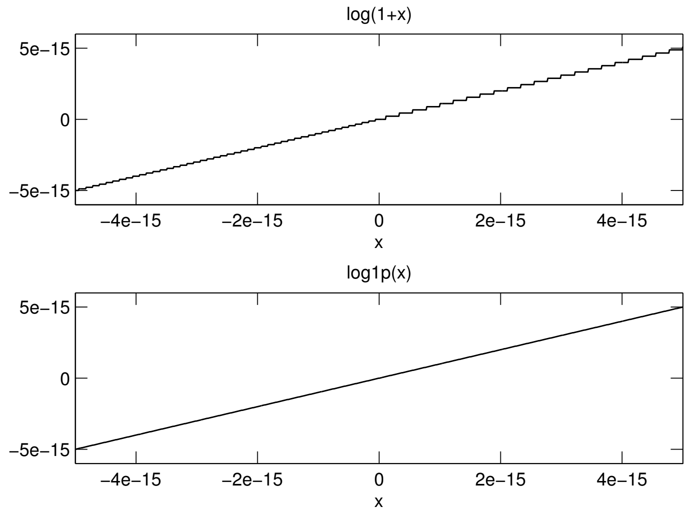

Figure A.1. $\log(1+x)$ *evaluated using MATLAB's* `log` (*top*) *and using the formula* (A.1) (*bottom*).

Dividing by $\epsilon(n-1)V(x)$ and taking the limit as $\epsilon \to 0$ gives the expression for κ_C. The result for κ_N follows from

$$\left|\sum_{j=1}^{n}(x_i - \overline{x})\Delta x_i\right| \le \left(\sum_{i=1}^{n}(x_i - \overline{x})^2\right)^{1/2} \epsilon\|x\|_2 = \epsilon(n-1)^{1/2}V(x)^{1/2}\|x\|_2,$$

where the inequality is attainable (Cauchy–Schwarz), together with the relation $(n-1)V(x) = \|x\|_2^2 - n\overline{x}^2$. That $\kappa_N \ge \kappa_C$ is easily verified.

1.8. The general solution to the recurrence is

$$x_k = \frac{100^{k+1}a + 6^{k+1}b + 5^{k+1}c}{100^k a + 6^k b + 5^k c},$$

where a, b, and c are arbitrary constants. The particular starting values chosen yield $a = 0$, $b = c = 1$, so that

$$x_k = \frac{6^{k+1} + 5^{k+1}}{6^k + 5^k} = 6 - \frac{1}{1 + (6/5)^k}.$$

Rounding errors in the evaluation (and in the representation of x_1 on a binary machine) cause a nonzero "a" term to be introduced, and the computed values are approximately of the form

$$\widehat{x}_k \approx \frac{100^{k+1}\eta + 6^{k+1} + 5^{k+1}}{100^k\eta + 6^k + 5^k},$$

for a constant η of order the unit roundoff. Hence the computed iterates rapidly converge to 100. Note that resorting to higher precision merely delays the inevitable convergence to 100. Priest [955, 1992, pp. 54–56] gives some interesting observations on the stability of the evaluation of the recurrence.

1.9. Writing

$$C := \mathrm{adj}(A) = \begin{bmatrix} a_{22} & -a_{12} \\ -a_{21} & a_{11} \end{bmatrix},$$

we have

$$\begin{aligned}
\widehat{x} &= fl\left(\frac{1}{d}Cb\right) \\
&= \frac{1}{d}(C + \Delta C)b, \qquad |\Delta C| \le \gamma_3 |C|, \\
&= x + \frac{1}{d}\Delta C b.
\end{aligned}$$

So

$$\frac{\|x - \widehat{x}\|_\infty}{\|x\|_\infty} \le \gamma_3 \frac{\|\,|A^{-1}||b|\,\|_\infty}{\|x\|_\infty} \le \gamma_3 \operatorname{cond}(A, x).$$

Also, $r = b - A\widehat{x} = -(1/d)A\Delta C b$, so $|r| \le \gamma_3 |A||A^{-1}||b|$, which implies the normwise residual bound. Note that the residual bound shows that $\|r\|_\infty / (\|A\|_\infty \|x\|_\infty)$ will be small if x is a large-normed solution ($\|x\|_\infty \approx \|A^{-1}\|_\infty \|b\|_\infty$).

1.10. Let $\widehat{m} = fl(\overline{x})$. For any standard summation method we have (see §4.2)

$$\begin{aligned}
\widehat{m} &= n^{-1} \sum_{i=1}^n x_i(1 + \delta_i), \qquad |\delta_i| \le \gamma_{n-1}, \\
&= m + \Delta m, \qquad \Delta m := n^{-1} \sum_{i=1}^n x_i \delta_i.
\end{aligned}$$

Then, since $fl\big((x_i - \widehat{m})^2\big) = (x_i - \widehat{m})^2 (1 + \alpha_i)^2 (1 + \beta_i)$, $|\alpha_i|, |\beta_i| \le u$, $\widehat{V}$ satisfies

$$\begin{aligned}
(n-1)\widehat{V} &= \sum_{i=1}^n (x_i - \widehat{m})^2 (1 + \epsilon_i), \qquad |\epsilon_i| \le \gamma_{n+3} \\
&= \sum_{i=1}^n \big[(x_i - m)^2 - 2\Delta m (x_i - m) + \Delta m^2\big](1 + \epsilon_i) \\
&= (n-1)V + \sum_{i=1}^n (x_i - m)^2 \epsilon_i \\
&\qquad + \sum_{i=1}^n (-2\Delta m(x_i - m) + \Delta m^2)(1 + \epsilon_i).
\end{aligned}$$

Hence, defining $\mu := n^{-1} \sum_{i=1}^n |x_i|$, so that $|\Delta m| \le \gamma_{n-1}\mu$,

$$\begin{aligned}
(n-1)|V - \widehat{V}| &\le \gamma_{n+3}(n-1)V + \sum_{i=1}^n |-2\Delta m(x_i - m)\epsilon_i + \Delta m^2 (1 + \epsilon_i)| \\
&\le \gamma_{n+3}(n-1)V + 2\gamma_{n-1}\gamma_{n+3}\mu\sqrt{n(n-1)V} + \gamma_{n-1}^2 \mu^2 n (1 + \gamma_{n+3}),
\end{aligned}$$

that is,

$$\begin{aligned}
\frac{|V - \widehat{V}|}{V} &\le \gamma_{n+3} + 2\theta\gamma_{n-1}\gamma_{n+3} + \theta^2 \gamma_{n-1}^2 (1 + \gamma_{n+3}) \\
&= (n+3)u + O(u^2),
\end{aligned}$$

where

$$\theta = \mu \sqrt{\frac{n}{(n-1)V}} \le \frac{\|x\|_2}{\sqrt{(n-1)V}} = \frac{1}{2}\kappa_N.$$

2.1. There are $1 + 2(e_{\max} - e_{\min} + 1)(\beta - 1)\beta^{t-1}$ normalized numbers (the "1" is for zero), and $2(\beta^{t-1} - 1)$ subnormal numbers. For IEEE arithmetic we therefore have

	Normalized	Subnormal
Single precision	4.3×10^9	1.7×10^7
Double precision	1.8×10^{19}	9×10^{15}

2.2. Without loss of generality suppose $x > 0$. We can write $x = m \times \beta^{e-t}$, where $\beta^{t-1} \le m < \beta^t$. The next larger floating point number is $x + \Delta x$, where $\Delta x = \beta^{e-t}$, and

$$\beta^{-1}\epsilon_M|x| = \beta^{-t}|x| < \frac{|x|}{m} = |\Delta x| \le \beta^{e-t}(\beta^{1-t}m) = \epsilon_M|x|.$$

The same upper bound clearly holds for the "next smaller" case, and the lower bound in this case is also easy to see.

2.3. The answer is the same for all adjacent nonzero pairs of single precision numbers. Suppose the numbers are 1 and $1 + \epsilon_M(\text{single}) = 1 + 2^{-23}$. The spacing of the double precision numbers on $[1, 2]$ is 2^{-52}, so the answer is $2^{29} - 1 \approx 5.4 \times 10^8$.

2.4. Inspecting the proof of Theorem 2.2 we see that $|y_i| \ge \beta^{e-1}$, $i = 1, 2$, and so we also have $|fl(x) - x|/|fl(x)| \le u$, that is, $fl(x) = x/(1 + \delta)$, $|\delta| \le u$. Note that this is not the same δ as in Theorem 2.2, although it has the same bound, and unlike in Theorem 2.2 there can be equality in this bound for δ.

2.5. The first part is trivial. Since, in binary notation,

$$x = 0.1100\ 1100\ 1100\ 1100\ 1100\ 1100 \mid 1100\ldots \times 2^{-3},$$

we have, rounding to 24 bits,

$$\widehat{x} = 0.1100\ 1100\ 1100\ 1100\ 1100\ 1101 \times 2^{-3}.$$

Thus

$$\begin{aligned}\widehat{x} - x &= (0.001 - 0.000\overline{1100})_2 \times 2^{-21} \times 2^{-3}\\ &= \left(\frac{1}{8} - \frac{1}{10}\right) \times 2^{-24} = \frac{1}{40} \times 2^{-24},\end{aligned}$$

and so $(x - \widehat{x})/x = -\frac{1}{4} \times 2^{-24} = -\frac{1}{4}u$.

2.6. Since the double precision significand contains 53 bits, $p = 2^{53} = 9007199254740992 \approx 9.01 \times 10^{15}$. For single precision, $p = 2^{24} = 16777216 \approx 1.68 \times 10^7$.

2.7.

1. True, since in IEEE arithmetic $fl(a \mathbin{\mathrm{op}} b)$ is defined to be the rounded value of $a \mathbin{\mathrm{op}} b$, $\mathrm{round}(a \mathbin{\mathrm{op}} b)$, which is the same as $\mathrm{round}(b \mathbin{\mathrm{op}} a)$.
2. True for round to nearest (the default) and round to zero, but false for round to $\pm\infty$.
3. True, because $fl(a + a) := \mathrm{round}(a + a) = \mathrm{round}(2 * a) =: fl(2 * a)$.
4. True: similar to 3.
5. False, in general.

6. True for binary arithmetic. Since the division by 2 is exact, the inequality is equivalent to $2a \le fl(a+b) \le 2b$. But $a+b \le 2b$, so, by the monotonicity of rounding, $fl(a+b) = \mathrm{round}(a+b) \le \mathrm{round}(2b) = 2b$. The lower bound is verified similarly.

2.8. Examples in 3-digit decimal arithmetic are $fl((5.01+5.02)/2) = fl(10.0/2) = 5.0$ and $fl((5.02+5.04)/2) = fl(10.1/2) = 5.05$.

The left-hand inequality is immediate, since $fl((b-a)/2) \ge 0$. The right-hand inequality can possibly be violated only if a and b are close (otherwise $(a+b)/2$ is safely less than b, so the inequality will hold). But then, by Theorem 2.5, $fl(b-a)$ is obtained exactly, and the result will have several zero low-order digits in the significand because of cancellation. Consequently, if the base is even the division is done exactly, and the right-hand inequality follows by the monotonicity of rounding. Alternatively, and even more simply, for any base we can argue that $fl((b-a)/2) \le b-a$, so that $a + fl((b-a)/2) \le b$ and, again, the result follows by the monotonicity of rounding.

2.9.

$$\begin{aligned}\sqrt{1-2^{-53}} &= 1 - \frac{1}{2}\cdot 2^{-53} - \frac{1}{8}\cdot 2^{-106} - \cdots \\ &= 0.\underbrace{11\ldots1}_{53}0\underbrace{11\ldots1}_{54}011\ldots \quad \text{(binary)}.\end{aligned}$$

Rounded directly to 53 bits this yields $1-2^{-53}$. But rounded first to 64 bits it yields

$$0.\underbrace{11\ldots1}_{53}1\underbrace{00\ldots0}_{10},$$

and when this number is rounded to 53 bits using the round to even rule it yields 1.0.

2.12. The spacing between the floating point numbers in the interval $(1/2, 1]$ is $\epsilon/2$ (cf. Lemma 2.1), so $|1/x - fl(1/x)| \le \epsilon/4$, which implies that $|1 - xfl(1/x)| \le \epsilon x/4 < \epsilon/2$. Thus

$$1 - \epsilon/2 < xfl(1/x) < 1 + \epsilon/2.$$

Since the spacing of the floating point numbers just to the right of 1 is ϵ, $xfl(1/x)$ must round to either $1-\epsilon/2$ or 1.

2.13. If there is no double rounding the answer is 257, 736, 490. For a proof that combines mathematical analysis with computer searching, see Edelman [379, 1994].

2.15. The IEEE standard does not define the results of exponentiation. The choice $0^0 = 1$ can be justified in several ways. For example, if $p(x) = \sum_{j=0}^{n} a_j x^j$ then $p(0) = a_0 = a_0 \times 0^0$, and the binomial expansion $(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}$ yields $1 = 0^0$ for $x = 0$, $y = 1$. For more detailed discussions see Goldberg [496, 1991, p. 32] and Knuth [742, 1992, pp. 406–408].

2.17. If the rounding mode is round to zero or round to $-\infty$ then $fl(2x_{\max}) = x_{\max}$.

2.18. No. A counterexample in 2-digit base 10 arithmetic is $fl(2.0 - 0.91) = fl(1.09) = 1.1$.

2.19. Every floating point number (whether normalized or unnormalized) is a multiple of $\mu = \beta^{e_{\min}-t}$ (although the converse is not true), therefore $x \pm y$ is also a multiple of μ. The subnormal numbers are equally spaced between μ and $\beta^{e_{\min}-1}$, the smallest positive normalized floating point number. But since $fl(x \pm y)$ underflows, $|x \pm y| < \beta^{e_{\min}-1}$, and therefore $x \pm y$ is exactly representable as a subnormal number. Thus $fl(x \pm y) = x \pm y$.

2.20. The function $fl(\sqrt{x})$ maps the set of positive floating point numbers onto a set of floating point numbers with about half as many elements. Hence there exist two distinct floating point numbers x having the same value of $fl(\sqrt{x})$, and so the condition $(\sqrt{x})^2 = |x|$ cannot always be satisfied in floating point arithmetic. The requirement $\sqrt{x^2} = |x|$ is reasonable for base 2, however, and is satisfied in IEEE arithmetic, as we now show.

Without loss of generality, we can assume that $1 \le x < 2$, since scaling x by a power of 2 does not alter $x/fl(\sqrt{x^2})$. By definition, $fl(\sqrt{x^2})$ is the nearest floating point number to $\sqrt{fl(x^2)} = \sqrt{x^2(1+\delta)}$, $|\delta| \le u$, and

$$\sqrt{fl(x^2)} = |x|(1+\delta)^{1/2} = |x|\left(1 + \frac{\delta}{\sqrt{1+\delta}+1}\right) = |x| + \theta.$$

Now the spacing of the floating point numbers between 1 and 2 is $\epsilon_M = 2u$, so

$$|\theta| \le (2-2u)\frac{u}{\sqrt{1-u}+1}.$$

Hence $|\theta| < u$ if $u \le 1/4$ (say), and then $|x|$ is the nearest floating point number to $\sqrt{fl(x^2)}$, so that $fl(\sqrt{fl(x^2)}) = |x|$.

In base 10 floating point arithmetic, the condition $\sqrt{x^2} = |x|$ can be violated. For example, working to 5 significant decimal digits, if $x = 3.1625$ then $fl(x^2) = fl(10.0014\,0625) = 10.001$, and $fl(\sqrt{10.001}) = fl(3.1624\,3577\ldots) = 3.1624 < x$.

2.21. On a Cray Y-MP the answer is yes, but in base 2 IEEE arithmetic the answer is no. It suffices to demonstrate that $fl(x/\sqrt{x^2}) = \mathrm{sign}(x)$ $(x \ne 0)$, which is shown by the proof of Problem 2.20.

2.22. The test "$x > y$" returns false if x or y is a NaN, so the code computes $\max(\mathrm{NaN}, 5) = 5$ and $\max(5, \mathrm{NaN}) = \mathrm{NaN}$, which violates the obvious requirement that $\max(x, y) = \max(y, x)$. Since the test $x \ne x$ identifies a NaN, the following code implements a reasonable definition of $\max(x, y)$:

```
% max(x, y)
if x ≠ x then
  max = y
else if y ≠ y then
      max = x
    else if y > x then
          max = y
        else
          max = x
        end
    end
end
```

A further refinement is to ensure that $\max(-0, +0) = +0$, which is not satisfied by the code above since -0 and $+0$ compare as equal; this requires bit-level programming.

2.23. We give an informal proof; the details are obtained by using the model $fl(x \,\mathrm{op}\, y) = (x \,\mathrm{op}\, y)(1+\delta)$, but they obscure the argument.

Since a, b, and c are nonnegative, $a + (b + c)$ is computed accurately. Since $c \le b \le a$, $c + (a - b)$ and $a + (b - c)$ are the sums of two positive numbers and so are computed accurately. Since a, b, and c are the lengths of sides of a triangle, $a \le b + c$; hence, using $c \le b \le a$,

$$b \le a \le b + c \le 2b,$$

which implies that $a - b$ is computed exactly, by Theorem 2.5. Hence $c - (a - b)$ is the difference of two exactly represented numbers and so is computed accurately. Thus $fl(A) = fl(\frac{1}{4}\sqrt{\widehat{x}})$, where $\widehat{x}$ is an accurate approximation to the desired argument x of the square root. It follows that $fl(A)$ is accurate.

2.24. For a machine with a guard digit, $y = x$, by Theorem 2.5 (assuming $2x$ does not overflow). If the machine lacks a guard digit then the subtraction produces x if the last bit of x is zero, otherwise it produces an adjacent floating point number with a zero last bit; in either case the result has a zero last bit. Gu, Demmel, and Dhillon [528, 1994] apply this bit zeroing technique to numbers $d_1, d_2, \ldots, d_n$ arising in a divide and conquer bidiagonal SVD algorithm, their motivation being that the differences $d_i - d_j$ can then be computed accurately even on machines without a guard digit.

2.25. The function $f(x) = 3x - 1$ has the single root $x = 1/3$. We can have $fl(f(x)) = 0$ only for $x \approx 1/3$. For $x \approx 1/3$ the evaluation can be summarized as follows:

$$\begin{aligned} x - 0.5 &\approx -\frac{1}{6}, \\ -\frac{1}{6} + x &\approx \frac{1}{6}, \\ \frac{1}{6} - 0.5 &\approx -\frac{1}{3}, \\ -\frac{1}{3} + x &\approx 0. \end{aligned}$$

The first, second, and fourth subtractions are done exactly, by Theorem 2.5. The result of the first subtraction has a zero least-significant bit and the result of the second has two zero least-significant bits; consequently the third subtraction suffers no loss of trailing bits and is done exactly. Therefore $f(x)$ is computed exactly for $x \equiv fl(x)$ near $1/3$. But $fl(x)$ can never equal $1/3$ on a binary machine, so $fl(f(x)) \ne 0$ for all x.

2.26. If x and y are t-digit floating point numbers then xy has $2t - 1$ or $2t$ significant digits. Since the error $fl(xy) - xy$ is less than one ulp in xy (i.e., beyond the tth significant digit), the difference $xy - fl(xy)$ is exactly representable in t digits, and it follows from the definition of FMA that the difference is computed exactly.

2.27. We have

$$\begin{aligned} \widehat{w} &= bc(1 + \delta_1), \\ \widehat{e} &= (\widehat{w} - bc)(1 + \delta_2), \\ (1 + \delta_4)\widehat{x} &= (ad - \widehat{w})(1 + \delta_3) + \widehat{e}, \end{aligned}$$

where $|\delta_i| \le u$, $i = 1\colon 4$. Hence

$$(1 + \delta_4)\widehat{x} = (ad - bc - bc\delta_1)(1 + \delta_3) + bc\delta_1(1 + \delta_2) = x + x\delta_3 - bc\delta_1(\delta_3 - \delta_2),$$

so that

$$|x - \widehat{x}| \le u(|x| + |\widehat{x}|) + 2u^2|bc|,$$

which implies high relative accuracy unless $u|bc| \gg |x|$. For comparison, the bound for standard evaluation of $fl(ad - bc)$ is $|x - \widehat{x}| \le \gamma_2(|ad| + |bc|)$.

2.28. We would like (2.8) to be satisfied, that is, we want $\widehat{z} = fl(x/y)$ to satisfy

$$\widehat{z} = (x/y)(1+\delta) + \eta, \qquad |\delta| \le u, \quad |\eta| \le \lambda u. \tag{A.2}$$

This implies that

$$|\widehat{z}y - x| = |x\delta + \eta y| \le u|x| + \lambda u|y|.$$

In place of $r = \widehat{z}y - x$ we have to use $\widehat{r} = fl(\widehat{z}y - x) = \widehat{z}y(1+\epsilon) - x$, where $|\epsilon| \le u$, and where we can assume the subtraction is exact, since the case of interest is where $\widehat{z}y \approx x$ (see Theorem 2.5). Thus $|\widehat{r}| = |r + \widehat{z}y\epsilon| \le |r| + u|\widehat{z}||y|$, and bounding $|\widehat{z}||y|$ using (A.2) we obtain

$$|\widehat{r}| \le |r| + u(1+u)|x| + \lambda u^2|y| \approx |r| + u|x|.$$

Therefore the convergence test is

$$|\widehat{r}| \le 2u|x| + \lambda u|y|.$$

Since λu underflows to zero it cannot be precomputed, and we should instead compute $\lambda(u|y|)$.

3.1. The proof is by induction. Inductive step: for $\rho_n = +1$,

$$\begin{aligned}
\prod_{i=1}^{n}(1+\delta_i) = (1+\delta_n)(1+\theta_{n-1}) &= 1+\theta_n, \\
\theta_n &= \delta_n + (1+\delta_n)\theta_{n-1}, \\
|\theta_n| &\le u + (1+u)\frac{(n-1)u}{1-(n-1)u} \\
&= \frac{u(1-(n-1)u) + (1+u)(n-1)u}{1-(n-1)u} \\
&= \frac{nu}{1-(n-1)u} \le \gamma_n.
\end{aligned}$$

For $\rho_n = -1$ we find, similarly, that

$$|\theta_n| \le \frac{nu - (n-1)u^2}{1 - nu + (n-1)u^2} \le \gamma_n.$$

3.2. This result can be proved by a modification of the proof of Lemma 3.1. But it follows immediately from the penultimate line of the proof of Lemma 3.4.

3.3. The computed iterates $\widehat{q}_k$ satisfy

$$(1+\delta_k)\widehat{q}_k = a_k + \frac{b_k}{\widehat{q}_{k+1}}(1+\epsilon_k), \qquad |\delta_k|, |\epsilon_k| \le u.$$

Defining $\widehat{q}_k = q_k + e_k$, we have

$$\begin{aligned}
q_k + e_k + \delta_k\widehat{q}_k &= a_k + \frac{b_k}{q_{k+1}+e_{k+1}} + \frac{b_k\epsilon_k}{\widehat{q}_{k+1}} \\
&= a_k + \frac{b_k}{q_{k+1}} - \frac{b_k e_{k+1}}{q_{k+1}(q_{k+1}+e_{k+1})} + \frac{b_k\epsilon_k}{\widehat{q}_{k+1}}.
\end{aligned}$$

This gives

$$|e_k| \le u|\widehat{q}_k| + \frac{|b_k||e_{k+1}|}{(|\widehat{q}_{k+1}| - |e_{k+1}|)|\widehat{q}_{k+1}|} + u\frac{|b_k|}{|\widehat{q}_{k+1}|}.$$

The running error bound μ can therefore be computed along with the continued fraction as follows:

$q_{n+1} = a_{n+1}$
$f_{n+1} = 0$
for $k = n\colon -1\colon 0$
    $r_k = b_k/q_{k+1}$
    $q_k = a_k + r_k$
    $f_k = |q_k| + |r_k| + |b_k| f_{k+1}/((|q_{k+1}| - u f_{k+1}|)|q_{k+1}|)$
end
$\mu = u f_0$

The error bound is valid provided that $|q_{k+1}| - u f_{k+1} > 0$ for all k. Otherwise a more sophisticated approach is necessary (for example, to handle the case where $q_{k+1} = 0$, $q_k = \infty$, and q_{k-1} is finite).

3.4. We prove just the division result and the last result. Let $(1+\theta_k)/(1+\theta_j) = 1+\alpha$. Then $\alpha = (\theta_k - \theta_j)/(1+\theta_j)$, so

$$\begin{aligned} |\alpha| \le \frac{\frac{ku}{1-ku} + \frac{ju}{1-ju}}{1 - \frac{ju}{1-ju}} &= \frac{(k+j)u - 2kju^2}{(1-2ju)(1-ku)} \\ &= \frac{(k+j)u - 2kju^2}{1-(2j+k)u + 2jku^2} \le \frac{(k+j)u}{1-(2j+k)u} \le \gamma_{k+2j}. \end{aligned}$$

However, it is easy to verify that, in fact,

$$\frac{(k+j)u - 2kju^2}{(1-2ju)(1-ku)} \le \frac{(k+j)u}{1-(k+j)u}$$

if $j \le k$.

For the last result,

$$\begin{aligned} \gamma_k + \gamma_j + \gamma_k\gamma_j &= \frac{ku(1-ju) + ju(1-ku) + kju^2}{(1-ku)(1-ju)} \\ &= \frac{(k+j)u - kju^2}{1-(k+j)u + kju^2} \\ &< \frac{(k+j)u}{1-(k+j)u} = \gamma_{k+j}. \end{aligned}$$

3.5. We have $fl(AB) = AB + \Delta C = (A + \Delta C B^{-1})B =: (A + \Delta A)B$, where $\Delta A = \Delta C B^{-1}$, and $|\Delta C| \le \gamma_n |A||B|$ by (3.13). Similarly, $fl(AB) = A(B + \Delta B)$, $|\Delta B| \le \gamma_n |A^{-1}||A||B|$.

3.6. We have

$$|R| = |A\Delta B + \Delta A B + \Delta A \Delta B| \le 2\epsilon EF + \epsilon^2 EF,$$

which implies

$$\epsilon^2 g_{ij} + 2\epsilon g_{ij} - |r_{ij}| \ge 0 \quad \text{for all } i, j.$$

Solving these inequalities for ϵ gives the required lower bound.

The definition is flawed in general as can be seen from rank considerations. For example if A and B are vectors and $\mathrm{rank}(C) \ne 1$, then no perturbations ΔA and ΔB exist to satisfy $C = (A+\Delta A)(B+\Delta B)$! Thus we have to use a mixed forward/backward stability definition in which we perturb C by at most ϵ as well as A and B.

3.7. Lemma 3.5 shows that (3.2) holds provided that each $(1+\delta)^k$ product is replaced by $(1+\epsilon)(1+\delta)^{k-1}$, where $|\epsilon| \le \sqrt{2}\gamma_2$, where the $1+\epsilon$ corresponds to a multiplication $x_i y_i$. Thus we have

$$\widehat{S}_n = \sum_{i=1}^{n} x_i y_i (1+\alpha_i), \quad \text{where} \quad 1+\alpha_i = (1+\theta_{n-1}^{(i)})(1+\epsilon_i).$$

It is easy to show that $|\alpha_i| \le \gamma_{n+2}$, so the only change required in (3.4) is to replace γ_n by γ_{n+2}. The complex analogue of (3.11) is $\widehat{y} = (A+\Delta A)x$, $|\Delta A| \le \gamma_{n+2}|A|$.

3.8. We have $fl((x+y)(x-y)) = (x+y)(x-y)(1+\theta_3)$, $|\theta_3| \le \gamma_3 = 3u/(1-3u)$, so the computed result has small relative error. Moreover, if $y/2 \le x \le y$ then $x-y$ is computed exactly, by Theorem 2.5, hence $fl((x+y)(x-y)) = (x+y)(x-y)(1+\theta_2)$. However, $fl(x^2-y^2) = x^2(1+\theta_2) - y^2(1+\theta_2')$, so that

$$\left|\frac{fl(x^2-y^2)-(x^2-y^2)}{x^2-y^2}\right| = \left|\frac{x^2\theta_2 - y^2\theta_2'}{x^2-y^2}\right| \le \gamma_2 \frac{x^2+y^2}{|x^2-y^2|},$$

and we cannot guarantee a small relative error.

If $|x| \gg |y|$ then $fl(x^2-y^2)$ suffers only two rounding errors, since the error in forming $fl(y^2)$ will not affect the final result, while $fl((x+y)(x-y))$ suffers three rounding errors; in this case $fl(x^2-y^2)$ is likely to be the more accurate result.

3.9. Assume the result is true for $m-1$. Now

$$\prod_{j=0}^{m}(X_j+\Delta X_j) = (X_m+\Delta X_m)\prod_{j=0}^{m-1}(X_j+\Delta X_j),$$

so

$$\begin{aligned}
\left\|\prod_{j=0}^{m}(X_j+\Delta X_j) - \prod_{j=0}^{m} X_j\right\| &= \left\| X_m\left[\prod_{j=0}^{m-1}(X_j+\Delta X_j) - \prod_{j=0}^{m-1} X_j\right]\right. \\
&\qquad \left. + \Delta X_m \prod_{j=0}^{m-1}(X_j+\Delta X_j)\right\| \\
&\le \|X_m\|\left(\prod_{j=0}^{m-1}(1+\delta_j) - 1\right)\prod_{j=0}^{m-1}\|X_j\| \\
&\qquad + \delta_m\|X_m\|_2 \prod_{j=0}^{m-1}\big(\|X_j\|(1+\delta_j)\big) \\
&= \left(\prod_{j=0}^{m}(1+\delta_j) - 1\right)\prod_{j=0}^{m}\|X_j\|.
\end{aligned}$$

3.10. Without loss of generality we can suppose that the columns of the product are computed one at a time. With $x_j = A_1 \dots A_k e_j$ we have, using (3.11),

$$fl(x_j) = (A_1+\Delta A_1)\dots(A_k+\Delta A_k)e_j, \qquad \|\Delta A_i\|_2 \le \sqrt{n}\gamma_n\|A_i\|_2,$$

and so, by Lemma 3.6,

$$\|x_j - fl(x_j)\|_2 \le \big[(1+\sqrt{n}\gamma_n)^k - 1\big]\|A_1\|_2 \dots \|A_k\|_2.$$

Squaring these inequalities and summing over j yields

$$\|A_1\dots A_k - fl(A_1\dots A_k)\|_F \le \sqrt{n}\big[(1+\sqrt{n}\gamma_n)^k - 1\big]\|A_1\|_2\dots\|A_k\|_2,$$

which gives the result.

3.11. The computations can be expressed as

$$\begin{aligned}
&y_1 = x^2\\
&y_{i+1} = \sqrt{y_i},\ i = 1{:}\,m\\
&z_1 = y_{m+1}\\
&z_{i+1} = z_i^2,\ i = 1{:}\,m-1
\end{aligned}$$

We have

$$\begin{aligned}
&\widehat{y}_1 = x^2(1+\delta_0)\\
&\widehat{y}_{i+1} = \sqrt{\widehat{y}_i}(1+\delta_i),\ i = 1{:}\,m\\
&\widehat{z}_1 = \widehat{y}_{m+1}\\
&\widehat{z}_{i+1} = \widehat{z}_i^{\,2}(1+\epsilon_i),\ i = 1{:}\,m-1
\end{aligned}$$

where $|\delta_i|, |\epsilon_i| \le u$. Solving these recurrences, we find that

$$\widehat{y}_{m+1} = (1+\delta_m)(1+\delta_{m-1})^{1/2}(1+\delta_{m-2})^{1/4}\dots(1+\delta_0)^{1/2^m}\,y_1^{1/2^m}.$$

It follows that

$$(1-u)^2 y_1^{\frac{1}{2^m}} \le \widehat{y}_{m+1} \le (1+u)^2 y_1^{\frac{1}{2^m}},$$

which shows that $\widehat{y}_{m+1}$ differs negligibly from y_{m+1}. For the repeated squarings, however, we find that

$$\begin{aligned}
\widehat{z}_m &= (1+\epsilon_{m-1})(1+\epsilon_{m-2})^2(1+\epsilon_{m-3})^4\dots(1+\epsilon_1)^{2^{m-1}}\widehat{z}_1^{\,2^{m-1}}\\
&= (1+\theta_{2^m-1})\widehat{z}_1^{\,2^{m-1}} = (1+\theta_{2^m+1})|x|,
\end{aligned}$$

where we have used Lemma 3.1. Hence the squarings introduce a relative error that can be approximately as large as $2^m u$. Since $u = 2^{-53}$, this relative error is of order 0.1 for $m = 50$, which explains the observed results for $m = 50$.

3.12. The analysis is just a slight extension of that for an inner product. The analogue of (3.3) is

$$\begin{aligned}
\widehat{J}(f) = w_1 f(x_1)&(1+\eta_1)(1+\theta_n) + w_2 f(x_2)(1+\eta_2)(1+\theta_n')\\
&+ w_3 f(x_3)(1+\eta_3)(1+\theta_{n-1}) + \dots + w_n f(x_n)(1+\eta_n)(1+\theta_2).
\end{aligned}$$

Hence

$$|I(f) - \widehat{J}(f)| \le (\eta + \gamma_n + \eta\gamma_n)\sum_{i=1}^{n}|w_i||f(x_i)|.$$

Setting $M = \max\{\,|f(x)| : a \le x \le b\,\}$, we have

$$|I(f) - \widehat{J}(f)| \le M(\eta + \gamma_n + \eta\gamma_n)\sum_{i=1}^{n}|w_i|. \tag{A.3}$$

Any reasonable quadrature rule designed for polynomic f has $\sum_{i=1}^n w_i = b - a$, so one implication of (A.3) is that it is best not to have weights of large magnitude and varying sign; ideally, $w_i \ge 0$ for all i (as for Gaussian integration rules, for example), so that $\sum_{i=1}^n |w_i| = b - a$.

4.1. A condition number is

$$C(x) = \max\left\{ \left|\frac{S_n(x) - S_n(x+\Delta x)}{\epsilon S_n(x)}\right| : |\Delta x| \le \epsilon|x| \right\}.$$

It is easy to show that

$$C(x) = \frac{S_n(|x|)}{|S_n(x)|} = \frac{\sum_{i=1}^n |x_i|}{\left|\sum_{i=1}^n x_i\right|}.$$

The condition number is 1 if the x_i all have the same sign.

4.2. In the $(i-1)$st floating point addition the "2^{k-t}" portion of x_i does not propagate into the sum (assuming that the floating point arithmetic uses round to nearest with ties broken by rounding to an even last bit or rounding away from zero), thus there is an error of 2^{k-t} and $\widehat{S}_i = i$. The total error is

$$2^{-t}(1 + 2^2 + 2^4 + \cdots + 2^{2(r-1)}) = 2^{-t}\frac{4^r - 1}{3},$$

while the upper bound of (4.4) is

$$(n-1)u\sum_{i=1}^{n}|x_i| + O(u^2) \le 2^r 2^{-t} 2^r + O(2^{-2t}) \approx 2^{-t}4^r,$$

which agrees with the actual error to within a factor 3; thus the smaller upper bound of (4.3) is also correct to within this factor. The example just quoted is, of course, a very special one, and as Wilkinson [1232, 1963, p. 20] explains, "in order to approach the upper bound as closely as this, not only must each error take its maximum value, but all the terms must be almost equal."

4.3. With $S_k = \sum_{i=1}^{k} x_i$, we have

$$\widehat{S}_k = fl(\widehat{S}_{k-1} + x_k) = (\widehat{S}_{k-1} + x_k)(1+\delta_k), \qquad |\delta_k| \le u, \quad k = 2\colon n.$$

By repeated use of this relation it follows that

$$\widehat{S}_n = (x_1 + x_2)\prod_{k=2}^{n}(1+\delta_k) + \sum_{i=3}^{n} x_i \prod_{k=i}^{n}(1+\delta_k),$$

which yields the required expression for $\widehat{S}_n$. The bound on $|E_n|$ is immediate.

The bound is minimized if the x_i are in increasing order of absolute value. This observation is common in the literature and it is sometimes used to conclude that the increasing ordering is the best one to use. This reasoning is fallacious, because minimizing an error bound is not the same as minimizing the error itself. As (4.3) shows, if we know nothing about the signs of the rounding errors then the "best" ordering to choose is one that minimizes the partial sums.

4.4. Any integer between 0 and 10 inclusive can be produced. For example, $fl(1+2+3+4+M-M) = 0$, $fl(M-M+1+2+3+4) = 10$, and $fl(2+3+M-M+1+4) = 5$.

4.5. This method is sometimes promoted on the basis of the argument that it minimizes the amount of cancellation in the computation of S_n. This is incorrect: the "$\pm$" method does not reduce the amount of cancellation—it simply concentrates all the cancellation into one step. Moreover, cancellation is not a bad thing per se, as explained in §1.7.

The "$\pm$" method is an instance of Algorithm 4.1 (assuming that S_+ and S_- are computed using Algorithm 4.1) and it is easy to see that it maximizes $\max_i |T_i|$ over all methods of this form (where, as in §4.2, T_i is the sum computed at the ith stage). Moreover, when $\sum_{i=1}^{n}|x_i| \gg |\sum_{i=1}^{n} x_i|$ the value of $\max_i |T_i|$ tends to be much larger for the "$\pm$" method than for the other methods we have considered.

4.6. Suppose, without loss of generality, that $|a| \ge |b|$. By definition, $fl(a+b)$ solves $\min|(a+b)-x|$ over all floating point numbers x. Since a is a floating point number,

$$\mathrm{err}(a,b) \le |(a+b)-a| = |b|,$$

which gives the first part. Now the least significant nonzero bit of $\mathrm{err}(a,b)$ is no smaller than 1 ulp in b (cf. Figure 4.1), and by the first part, $|\mathrm{err}(a,b)| \le |b|$. Hence the significand of $\mathrm{err}(a,b)$ is no longer than that of b, which means that it is representable as a floating point number in the absence of overflow.

4.7. The main concern is to evaluate the denominator accurately when the x_i are close to convergence. The bound (4.3) tells us to minimize the partial sums; these are, approximately, for $x_i \approx \xi$, (a) $-\xi$, 0, (b) 0, 0, (c) 2ξ, 0. Hence the error analysis of summation suggests that (b) is the best expression, with (a) a distant second. That (b) is the best choice is confirmed by Theorem 2.5, which shows there will be only one rounding error when the x_i are close to convergence. A further reason to prefer (b) is that it is less prone to overflow than (a) and (c).

4.8. This is, of course, not a practical method, not least because it is very prone to overflow and underflow. However, its error analysis is interesting. Ignoring the error in the log evaluation, and assuming that exp is evaluated with relative error bounded by u, we have, with $|\delta_i| \le u$ for all i, and for some δ_{2n}

$$\begin{aligned}\widehat{S}_n &= \log\bigl(e^{x_1}\dots e^{x_n}(1+\delta_1)\dots(1+\delta_{2n-1})\bigr)\\ &= \log\bigl(e^{x_1}\dots e^{x_n}(1+\delta_{2n})^{2n-1}\bigr)\\ &= x_1+\cdots+x_n+(2n-1)\log(1+\delta_{2n})\\ &\approx x_1+\cdots+x_n+(2n-1)\delta_{2n}.\end{aligned}$$

Hence the best relative error bound we can obtain is $|S_n - \widehat{S}_n|/|S_n| \lesssim (2n-1)u/|S_n|$. Clearly, this method of evaluation guarantees a small absolute error, but not a small relative error when $|S_n| \ll 1$.

4.9. Method (a) is recursive summation of $a, h, h, \dots, h$. From (4.3) we have $|a + i\widehat{h} - \widehat{x}_i| \le [ia + \frac{1}{2}i(i+1)\widehat{h}]u + O(u^2)$. Hence, since $\widehat{h} = h(1+\epsilon)$, $|\epsilon| \le u$,

$$\begin{aligned}|x_i - \widehat{x}_i| &\le |ih - i\widehat{h}| + |a + i\widehat{h} - \widehat{x}_i| \le ihu + i(|a| + \tfrac{1}{2}(i+1)h)u + O(u^2)\\ &= i(|a| + \tfrac{1}{2}(i+3)h)u + O(u^2).\end{aligned}$$

For (b), using the relative error counter notation (3.10), $\widehat{x}_i = (a + i\widehat{h}<1>)<1> = a<1> + ih<3>$. Hence

$$|x_i - \widehat{x}_i| \le u|a| + \gamma_3 ih.$$

For (c), $\widehat{x}_i = a(1 - <1> i/n)<2> + (i/n)b<3>$, hence

$$|x_i - \widehat{x}_i| \le \gamma_2|a| + \gamma_3|a|i/n + \gamma_3|b|i/n \le \gamma_3\bigl(|a|(1+i/n) + |b|i/n\bigr).$$

The error bound for (b) is about a factor i smaller than that for (a). Note that method (c) is the only one guaranteed to yield $\widehat{x}_n = b$ (assuming $fl(n/n) = 1$, as holds in IEEE arithmetic), which may be important when integrating a differential equation to a given end-point.

If $a \ge 0$ then the bounds imply

$$\begin{aligned}(a):&\quad |x_i - \widehat{x}_i| \lesssim iu|x_i|,\\ (b):&\quad |x_i - \widehat{x}_i| \le \gamma_3|x_i|,\\ (c):&\quad |x_i - \widehat{x}_i| \le \gamma_3(a + i/n(a+b)) = \gamma_3(a + i/n(b - a + 2a))\\ &\qquad\qquad = \gamma_3(a + ih + 2ai/n) \le 3\gamma_3|x_i|.\end{aligned}$$

Thus (b) and (c) provide high relative accuracy for all i, while the relative accuracy of (a) can be expected to degrade as i increases.

5.1. By differentiating the Horner recurrence $q_i = xq_{i+1} + a_i$, $q_n = a_n$, we obtain

$$\begin{aligned}
q_i' &= xq_{i+1}' + q_{i+1}, & q_n' &= 0, & q_{n-1}' &= a_n,\\
q_i'' &= xq_{i+1}'' + 2q_{i+1}', & q_n'' &= q_{n-1}'' = 0, & q_{n-2}'' &= 2a_n,\\
q_i''' &= xq_{i+1}''' + 3q_{i+1}'', & q_n''' &= q_{n-1}''' = q_{n-2}''' = 0, & q_{n-3}''' &= 6a_n.
\end{aligned}$$

The factors 2, 3, ..., can be removed by redefining $r_i^{(k)} = q_i^{(k)}/k!$. Then $r_i^{(k)} = xr_{i+1}^{(k)} + r_{i+1}^{(k-1)}$, $r_{n-k}^{(k)} = a_n$.

5.2. Analysis similar to that for Horner's rule shows that

$$fl(p(x)) = a_0\langle n\rangle + a_1x\langle n+1\rangle + \cdots + a_nx^n\langle n+1\rangle.$$

The total number of rounding errors is the same as for Horner's algorithm, but they are distributed more equally among the terms of the polynomial. Horner's rule can be expected to be more accurate when the terms $|a_ix^i|$ decrease rapidly with i, such as when $p(x)$ is the truncation of a rapidly convergent power series. Of course, this algorithm requires twice as many multiplications as Horner's method.

5.3. Accounting for the error in forming y, we have, using the relative error counter notation (3.10),

$$\begin{aligned}
fl(p(x)) &= \big[a_0\langle 1\rangle + a_2x^2\langle 4\rangle + \cdots + a_{n-2}x^{n-2}\langle 3m-2\rangle + a_nx^n\langle 3m\rangle\\
&\qquad + x\langle 1\rangle\big(a_1\langle 1\rangle + a_3x^2\langle 4\rangle + \cdots + a_{2n-1}x^{2n-1}\langle 3m-3\rangle\big)\big]\langle 1\rangle\\
&= \sum_{i=0}^{n} a_ix^i(1+\theta_{3m+1}^{(i)}), \qquad |\theta_{3m+1}^{(i)}| \le \gamma_{3m+1}.
\end{aligned}$$

Thus the relative backward perturbations are bounded by $(3n/2+1)u$ instead of $2nu$ for Horner's rule.

5.4. Here is a MATLAB M-file to perform the task.

```
function [a, perm] = leja(a)
%LEJA    LEJA ordering.
%        [A, PERM] = LEJA(A) reorders the points A by the
%        Leja ordering and returns the permutation vector that
%        effects the ordering in PERM.

n = max(size(a));
perm = (1:n)';

% a(1) = max(abs(a)).
[t, i] = max(abs(a));
if i ~= 1
   a([1 i]) = a([i 1]);
   perm([1 i]) = perm([i 1]);
end

p = ones(n,1);
for k=2:n-1
    for i=k:n
```

```
        p(i) = p(i)*(a(i)-a(k-1));
    end
    [t, i] = max(abs(p(k:n)));
    i = i+k-1;
    if i ~= k
       a([k i]) = a([i k]);
       p([k i]) = p([i k]);
       perm([k i]) = perm([i k]);
    end
end
```

5.5. It is easy to show that the computed $\widehat{p}$ satisfies $\widehat{p} = p(x)(1+\theta_{2n+1})$, $|\theta_{2n+1}| \le \gamma_{2n+1}$. Thus $\widehat{p}$ has a tiny relative error. Of course, this assumes that the roots x_i are known exactly!

6.1. For $\alpha_{S,2}$, if $A \in \mathbb{C}^{m\times n}$ then, using the Cauchy–Schwarz inequality,

$$\|A\|_S^2 = \Big(\sum_{i,j} |a_{ij}|\Big)^2 \le mn\|A\|_F^2 \le mn\,\mathrm{rank}(A)\|A\|_2^2.$$

The first inequality is an equality iff $|a_{ij}| \equiv \alpha$, and the second inequality is an equality iff A is a multiple of a matrix with orthonormal columns. If A is real and square, these requirements are equivalent to A being a scalar multiple of a Hadamard matrix. If A is complex and square, the requirements are satisfied by the given Vandermonde matrix, which is $\sqrt{n}$ times a unitary matrix.

6.2. $\|xy^*\| = \max_{\|z\|=1} \|xy^*z\| = \|x\| \max_{\|z\|=1} |y^*z| = \|x\|\,\|y\|_D$.

6.3. By the Hölder inequality,

$$\mathrm{Re}\, y^*Ax \le |y^*Ax| \le \|y\|_D\|A\|\,\|x\|. \tag{A.4}$$

We now show that equality is possible throughout (A.4). Let x satisfy $\|A\| = \|Ax\|/\|x\|$ and let y be dual to Ax. Then

$$\mathrm{Re}\, y^*Ax = y^*Ax = \|y\|_D\|Ax\| = \|y\|_D\|A\|\,\|x\|,$$

as required. The last part follows by applying this equality to A^* and using the fact that the dual of the dual vector norm is the original norm.

6.4. From (6.19) we have $\|M_n\|_p \le \mu_n^{1/p}\mu_n^{1-1/p} = \mu_n$. But by taking x in the definition (6.11) to be the vector of all ones, we see that $\|M_n\|_p \ge \mu_n$.

6.5. If $A = PDQ^*$ is an SVD then

$$\begin{aligned}\|AB\|_F &= \|PDQ^*B\|_F = \|DQ^*B\|_F\\ &\le \|\,\mathrm{diag}(\max_i \sigma_i)Q^*B\|_F = (\max_i \sigma_i)\|Q^*B\|_F\\ &= \|A\|_2\|B\|_F.\end{aligned}$$

Similarly, $\|BC\|_F \le \|B\|_F\|C\|_2$, and these two inequalities together imply the required one.

6.6. By (6.6) and (6.8) it suffices to show that $\|A^{-1}\|_{\beta,\alpha} = (\min_{x\neq 0} \|Ax\|_\beta/\|x\|_\alpha)^{-1}$. We have

$$\begin{aligned} \|A^{-1}\|_{\beta,\alpha} &= \max_{x\neq 0} \frac{\|A^{-1}x\|_\alpha}{\|x\|_\beta} \\ &= \max_{y\neq 0} \frac{\|y\|_\alpha}{\|Ay\|_\beta} \quad (y = A^{-1}x) \\ &= (\min_{x\neq 0} \|Ax\|_\beta/\|x\|_\alpha)^{-1}. \end{aligned}$$

6.7. Let λ be an eigenvalue of A and x the corresponding eigenvector, and form the matrix $X = [x, x, \dots, x] \in \mathbb{C}^{n\times n}$. Then $AX = \lambda X$, so $|\lambda|\|X\| = \|AX\| \le \|A\|\,\|X\|$, showing that $|\lambda| \le \|A\|$. For a subordinate norm it suffices to take norms in the equation $Ax = \lambda x$.

6.8. The following proof is much simpler than the usual proof based on diagonal scaling to make the off-diagonal of the Jordan matrix small (see, e.g., Horn and Johnson [636, 1985, Lem. 5.6.10]). The proof is from Ostrowski [909, 1973, Thm. 19.3].

Let $\delta^{-1}A$ have the Jordan canonical form $\delta^{-1}A = XJX^{-1}$. We can write $J = \delta^{-1}D + N$, where $D = \mathrm{diag}(\lambda_i)$ and the λ_i are the eigenvalues of A. Then $A = X(D + \delta N)X^{-1}$, so

$$\|A\| := \|X^{-1}AX\|_\infty = \|D + \delta N\|_\infty \le \rho(A) + \delta.$$

Note that we actually have $\|A\| = \rho(A)+\delta$ if the largest eigenvalue occurs in a Jordan block of size greater than 1. If A is diagonalizable then with $\delta = 0$ we get $\|A\| = \rho(A)$. The last part of the result is trivial.

6.9. We have $\|A\|_2 \le \|A\|_F \le \sqrt{n}\|A\|_2$ (in fact, we can replace $\sqrt{n}$ by $\sqrt{r}$, where $r = \mathrm{rank}(A)$). There is equality on the left when $\sigma_2 = \cdots = \sigma_n = 0$, that is, when A has rank 1 ($A = xy^*$) or $A = 0$. There is equality on the right when $\sigma_1 = \cdots = \sigma_n = \alpha$, that is, when $A = \alpha Q$ where Q has orthonormal columns, $\alpha \in \mathbb{C}$.

6.10. Let $F = P\Sigma Q^*$ be an SVD. Then

$$\left\| \begin{bmatrix} I & F \\ 0 & I \end{bmatrix} \right\|_2 = \left\| \begin{bmatrix} P & 0 \\ 0 & Q \end{bmatrix} \begin{bmatrix} I & \Sigma \\ 0 & I \end{bmatrix} \begin{bmatrix} P^* & 0 \\ 0 & Q^* \end{bmatrix} \right\|_2 = \left\| \begin{bmatrix} I & \Sigma \\ 0 & I \end{bmatrix} \right\|_2.$$

But $\left[\begin{smallmatrix} I & \Sigma \\ 0 & I \end{smallmatrix}\right]$ can be permuted into the form $\mathrm{diag}(D_i)$, where $D_i = \left[\begin{smallmatrix} 1 & \sigma_i \\ 0 & 1 \end{smallmatrix}\right]$. It is easy to find the singular values of D_i, and the maximum value is attained for $\sigma_1 = \|F\|_2$.

6.11. (a)

$$\|Ax\|_\beta = \Big\| \sum_j x_j A(:,j) \Big\|_\beta \le \max_j \|A(:,j)\|_\beta \sum_j |x_j|,$$

with equality for $x = e_k$, where the maximum is attained for $j = k$.

(b)

$$\|Ax\|_\infty = \max_i |A(i,:)x| \le \max_i \big(\|A(i,:)^*\|_\alpha^D \|x\|_\alpha \big) = \|x\|_\alpha \max_i \|A(i,:)^*\|_\alpha^D.$$

Equality is attained for an x that gives equality in the Hölder inequality involving the kth row of A, where the maximum is attained for $i = k$. Finally, from either formula, $\|A\|_{1,\infty} = \max_{i,j} |a_{ij}|$.

6.12. Using the Cholesky factorization $A = R^*R$,

$$\begin{aligned}\|A\|_{\infty,1} &= \max_{x\neq 0}\frac{\|Ax\|_1}{\|x\|_\infty} = \max\{\, y^*Ax : \|x\|_\infty = 1,\ \|y\|_\infty = 1\,\}\\ &= \max\{\,(Ry)^*Rx : \|x\|_\infty = 1,\ \|y\|_\infty = 1\,\}\\ &= \max\{\,(Rx)^*Rx : \|x\|_\infty = 1\,\}\\ &= \max\{\, x^*Ax : \|x\|_\infty = 1\,\}.\end{aligned}$$

6.13. H^T is also a Hadamard matrix, so by the duality result (6.21) it suffices to show that $\|H\|_p = n^{1/p}$ for $1 \le p \le 2$. Since $|h_{ij}| \equiv 1$, (6.12) gives $\|H\|_p \ge n^{1/p}$. Since $\|H\|_1 = n$ and $\|H\|_2 = n^{1/2}$, (6.20) gives $\|H\|_p \le n^{1/p}$, and so $\|H\|_p = n^{1/p}$ for $1 \le p \le 2$, as required.

6.14. We prove the lower bound in (6.23) and the upper bound in (6.24); the other bounds follow on using $\|A^T\|_p = \|A\|_q$. The lower bound in (6.23) is just (6.12). Now assume that A has at most μ nonzeros per column. Define

$$D_i = \mathrm{diag}(s_{i1},\dots,s_{in}), \qquad s_{ij} = \begin{cases}1, & a_{ij}\neq 0,\\ 0, & a_{ij} = 0,\end{cases}$$

and note that

$$\sum_i \|D_i x\|_p^p = \sum_{i,j} s_{ij}|x_j|^p = \sum_j |x_j|^p \sum_i s_{ij} \le \mu\|x\|_p^p.$$

We have

$$\begin{aligned}\|Ax\|_p^p &= \sum_i |A(i,:)x|^p = \sum_i |A(i,:)D_i x|^p\\ &\le \sum_i \|A(i,:)\|_q^p \|D_i x\|_p^p\\ &\le \max_i \|A(i,:)\|_q^p \sum_i \|D_i x\|_p^p\\ &\le \max_i \|A(i,:)\|_q^p\, \mu\|x\|_p^p,\end{aligned}$$

which gives the upper bound in (6.24).

6.15. The lower bound follows from $\|Ax\|_p/\|x\|_p \le \|\,|A||x|\,\|_p/\|\,|x|\,\|_p$. From (6.12) we have

$$\|\,|A|\,\|_p \le n^{1-1/p}\max_j \|\,|A|e_j\|_p = n^{1-1/p}\max_j \|Ae_j\|_p \le n^{1-1/p}\|A\|_p.$$

By (6.21), we also have $\|\,|A|\,\|_p = \|\,|A^T|\,\|_q \le n^{1-1/q}\|A^T\|_q = n^{1/p}\|A\|_p$ and the result follows.

6.16. The function ν is not a vector norm because $\nu(\alpha x) \neq |\alpha|\nu(x)$ does not hold for all $\alpha \in \mathbb{C}$, $x \in \mathbb{C}^n$. However, $\nu(\alpha x) \le \nu(\alpha)\nu(x)$, and the other two norm conditions hold, so it makes sense to define the "subordinate norm". We have

$$\begin{aligned}\nu(Ax) &= \nu\Big(\sum_j x_j A(:,j)\Big) \le \sum_j \nu(x_j A(:,j))\\ &\le \sum_j \nu(x_j)\nu(A(:,j)) \le \max_j \nu(A(:,j)) \sum_j \nu(x_j)\\ &= \max_j \nu(A(:,j))\nu(x).\end{aligned}$$

There is equality for x_j an appropriate unit vector e_j. Hence $\nu(A) = \max_j \nu(A(:,j))$.

7.1. It is straightforward to obtain from $A(y-x) = \Delta b - \Delta A x + \Delta A(x-y)$ the inequality $(I - \epsilon|A^{-1}|E)|x-y| \le \epsilon|A^{-1}|(f + E|x|)$. In general, if $B \ge 0$ and $\rho(B) < 1$ then $I - B$ is nonsingular. Since $\rho(\epsilon|A^{-1}|E) \le \epsilon\| \, |A^{-1}|E \, \| < 1$, we can premultiply by $(I - \epsilon|A^{-1}|E)^{-1} \ge 0$ to obtain the bound for $|x-y|$. For the last part,

$$\frac{|x_i - y_i|}{|x_i|} \le \frac{\epsilon\big(|A^{-1}|(f + E|x|)\big)_i}{|x_i|} + O(\epsilon^2).$$

7.2. Take norms in $r = A(x-y)$ and $x - y = A^{-1}r$. The result says that the normwise relative error is at least as large as the normwise relative residual and possibly $\kappa(A)$ times as large. Since the upper bound is attainable, the relative residual is not a good predictor of the relative error unless A is very well conditioned.

7.3. Let D_R equilibrate the rows of A, so that $B = D_R A$ satisfies $|B|e = e$. Then

$$\operatorname{cond}(B) = \| |B^{-1}||B|e \|_\infty = \| |B^{-1}|e \|_\infty = \|B^{-1}\|_\infty = \kappa_\infty(B).$$

Hence $\kappa_\infty(D_R A) = \operatorname{cond}(D_R A) = \operatorname{cond}(A)$, which implies $\kappa_\infty(A) \le \kappa_\infty(D_R)\operatorname{cond}(A)$. The inequality $\operatorname{cond}(A) \le \kappa_\infty(A)$ is trivial, and the deduction of (7.15) is immediate.

7.4. The first inequality is trivial. For the second, since $h_{ii} \equiv 1$ and $|h_{ij}| \le 1$ we have $|H| \ge I$ and $\|H\|_\infty \le n$. Hence

$$\| \, |H^{-1}||H| \, \|_\infty \ge \| \, |H^{-1}| \, \|_\infty = \|H^{-1}\|_\infty \ge \|H^{-1}\|_\infty \frac{\|H\|_\infty}{n} = \frac{\kappa_\infty(H)}{n}.$$

7.5. We have $\Delta x = A^{-1}\Delta b$, which yields $\|\Delta x\|_2 \le \sigma_n^{-1}\|\Delta b\|_2$. Now

$$\begin{aligned}
\|x\|_2^2 = \|V\Sigma^{-1}U^T b\|_2 &= \sum_{i=1}^n \frac{(u_i^T b)^2}{\sigma_i^2} \qquad (u_i = U(:,i)) \\
&\ge \sum_{i=n+1-k}^n \frac{(u_i^T b)^2}{\sigma_i^2} \\
&\ge \frac{1}{\sigma_{n+1-k}^2} \sum_{i=n+1-k}^n (u_i^T b)^2 = \frac{\|P_k b\|_2^2}{\sigma_{n+1-k}^2},
\end{aligned}$$

which yields the result.

If we take $k = n$ we obtain the bound that would be obtained by applying standard perturbation theory (Theorem 7.2). The gist of this result is that the full $\kappa_2(A)$ magnification of the perturbation will not be felt if b contains a significant component in the subspace $\operatorname{span}(U_k)$ for a k such that $\sigma_{n+1-k} \approx \sigma_n$. This latter condition says that x must be a large-normed solution: $\|x\|_2 \approx \|A^{-1}\|_2\|b\|_2$.

7.6. (a) Use $\| \, |A^{-1}|E|x| \, \|_\infty = \|x\|_1 \| \, |A^{-1}||A|e \, \|_\infty = \|x\|_1 \| \, |A^{-1}||A| \, \|_\infty$. For the upper bound, use $f \le |A||x|$.

(b) Use $\| \, |A^{-1}|E|x| \, \|_\infty = \| \, |A^{-1}|ee^T|A||x| \, \|_\infty = \|A^{-1}\|_\infty \| \, |A||x| \, \|_1$.

7.7. We will prove the result for ω; the proof for η is entirely analogous. The lower bound is trivial. Let $\epsilon = \omega_{|A|,|b|}(y)$ and $r = b - Ay$. Then $|r| \le \epsilon(|A||y| + |b|)$ and $(A + \Delta A)y = b + \Delta b$ with $|\Delta A| \le \epsilon|A|$ and $|\Delta b| \le \epsilon|b|$. Hence $|b| = |(A + \Delta A)y - \Delta b| \le (1+\epsilon)|A||y| + \epsilon|b|$, yielding $|b| \le (1-\epsilon)^{-1}(1+\epsilon)|A||y|$. Thus

$$|r| \le \epsilon\left(1 + \frac{1+\epsilon}{1-\epsilon}\right)|A||y| = \frac{2\epsilon}{1-\epsilon}|A||y|,$$

which implies the result, by Theorem 7.3.

7.8. The constraint on ΔA and Δb can be written

$$[\,\Delta A \quad \theta\Delta b\,]\begin{bmatrix} y \\ -\theta^{-1} \end{bmatrix} = r,$$

and so

$$\|r\|_2 \le (\|y\|_2^2 + \theta^{-2})^{1/2}\|[\Delta A,\ \theta\Delta b]\|_2 \le \theta^{-1}(\theta^2\|y\|_2^2 + 1)^{1/2}\|[\Delta A,\ \theta\Delta b]\|_F,$$

which gives a lower bound on $\eta_F(y)$. The bound is attained for the rank-1 perturbation $[\Delta A,\ \theta\Delta b] = r\,[y^T, -\theta^{-1}]^T/(\|y\|_2^2 + \theta^{-2})$.

7.9. We have $\Delta x = A^{-1}(\Delta b - \Delta Ax) + O(\epsilon^2)$. Therefore $c^T\Delta x = c^TA^{-1}(\Delta b - \Delta Ax) + O(\epsilon^2)$, and so

$$\frac{|c^T\Delta x|}{\epsilon|c^Tx|} \le \frac{|c^TA^{-1}|(f + E|x|)}{|c^Tx|} + O(\epsilon),$$

this inequality being sharp. Hence

$$\chi_{E,f}(A,x) = \frac{|c^TA^{-1}|(f + E|x|)}{|c^Tx|}.$$

The lower bound for $\chi_{|A|,|b|}(A,x)$ follows from the inequalities $|c^Tx| = |c^TA^{-1}\cdot Ax| \le |c^TA^{-1}|\,|A|\,|x|$ and $|c^Tx| = |c^TA^{-1}b| \le |c^TA^{-1}||b|$. A slight modification to the derivation of $\chi_{E,f}(A,x)$ yields

$$\psi_{E,f}(A,x) = \|c^TA^{-1}\|_2\frac{\|f\|_2 + \|E\|_2\|x\|_2}{|c^Tx|}.$$

7.10. (a) For any $D_1, D_2 \in \mathcal{D}_n$ we have

$$\rho(BC) = \rho(D_1BCD_1^{-1}) \le \|D_1BCD_1^{-1}\|_\infty \le \|D_1BD_2\|_\infty\|D_2^{-1}CD_1^{-1}\|_\infty. \tag{A.5}$$

The rest of the proof shows that equality can be attained in this inequality.

Let $x_1 > 0$ be a right Perron vector of BC, so that $BCx_1 = \pi x_1$, where $\pi = \rho(BC) > 0$. Define

$$x_2 = Cx_1, \tag{A.6}$$

so that $x_2 > 0$ and

$$Bx_2 = \pi x_1. \tag{A.7}$$

(We note, incidentally, that x_2 is a right Perron vector of CB: $CBx_2 = \pi x_2$.)

Now define

$$D_1 = \mathrm{diag}(x_1)^{-1}, \qquad D_2 = \mathrm{diag}(x_2). \tag{A.8}$$

Then, with $e = [1,1,\dots,1]^T$, (A.7) can be written $BD_2e = \pi D_1^{-1}e$, or $D_1BD_2e = \pi e$. Since $D_1BD_2 > 0$, this gives $\|D_1BD_2\|_\infty = \pi$. Similarly, (A.6) can be written $D_2e = CD_1^{-1}e$, or $D_2^{-1}CD_1^{-1}e = e$, which gives $\|D_2^{-1}CD_1^{-1}\|_\infty = 1$. Hence for D_1 and D_2 defined by (A.8) we have $\|D_1BD_2\|_\infty\|D_2^{-1}CD_1^{-1}\|_\infty = \pi = \rho(BC)$, as required.

Note that for the optimal D_1 and D_2, D_1BD_2 and $D_2^{-1}CD_1^{-1}$ both have the property that all their rows sums are equal.

(b) Take $B = |A|$ and $C = |A^{-1}|$, and note that $\kappa_\infty(D_1AD_2) = \|D_1|A|D_2\|_\infty \times \|D_2^{-1}|A^{-1}|D_1^{-1}\|_\infty$. Now apply (a).

(c) We can choose $F_1 > 0$ and $F_2 > 0$ so that $|A| + tF_1 > 0$ and $|A^{-1}| + tF_2 > 0$ for all $t > 0$. Hence, using (a),

$$\begin{aligned}\inf_{D_1,D_2\in\mathcal{D}_n}\kappa_\infty(D_1AD_2) &\le \inf_{D_1,D_2\in\mathcal{D}_n}\|D_1(|A| + tF_1)D_2\|_\infty\|D_2^{-1}(|A^{-1}| + tF_2)D_1^{-1}\|_\infty \\ &= \rho\big((|A| + tF_1)(|A^{-1}| + tF_2)\big).\end{aligned}$$

Taking the limit as $t \to 0$ yields the result, using continuity of eigenvalues.

(d) A nonnegative irreducible matrix has a positive Perron vector, from standard Perron–Frobenius theory (see, e.g., Horn and Johnson [636, 1985, Chap. 8]). Therefore the result follows by noting that in the proof of (a) all we need is for D_1 and D_2 in (A.8) to be defined and nonsingular, that is, for x_1 and x_2 to be positive vectors. This is the case if BC and CB are irreducible (since x_2 is a Perron vector of CB)

(e) Using $\kappa_1(A) = \kappa_\infty(A^T)$, $\rho(A) = \rho(A^T)$, and $\rho(AB) = \rho(BA)$, it is easy to show that the results of (a)–(d) remain true with the ∞-norm replaced by the 1-norm. From $\|A\|_2 \le \left(\|A\|_1\|A\|_\infty\right)^{1/2}$ it then follows that $\inf_{D_1,D_2\in\mathcal{D}_n} \kappa_2(D_1AD_2) \le \rho(|A|\,|A^{-1}|)$. In fact, the result in (a) holds for any p-norm, though the optimal D_1 and D_2 depend on the norm; see Bauer [90, 1963, Lem. 1(i)].

7.11. That $\mu_E'(A)$ cannot exceed the claimed expression follows by taking absolute values in the expression $\Delta X = -(A^{-1}\Delta A A^{-1} + A^{-1}\Delta A \Delta X)$. To show it can equal it we need to show that if the maximum is attained for $(i,j) = (r,s)$ then

$$|\Delta x_{rs}| = \epsilon\left(|A^{-1}|E|A^{-1}|\right)_{rs}$$

can be attained, to first order in ϵ. Equality is attained for $\Delta A = D_1ED_2$, where $D_1 = \mathrm{diag}(\mathrm{sign}(A^{-1})_{ri})$, $D_2 = \mathrm{diag}(\mathrm{sign}(A^{-1})_{is})$.

7.12. (a) We need to find a symmetric H satisfying $Hy = b - Ay =: r$. Consider the QR factorization

$$Q^T\,[\,y \quad r\,] = \begin{bmatrix} t_{11} & t_{12} \\ 0 & t_{22} \\ \vdots & \vdots \\ 0 & 0 \end{bmatrix}.$$

The constraint can be rewritten $Q^THQ \cdot Q^Ty = Q^Tr$, and it is satisfied by $Q^THQ := \mathrm{diag}(\widetilde{H}, 0_{n-2})$ if we can find $\widetilde{H} \in \mathbb{R}^{2\times 2}$ such that

$$\widetilde{H}t = u, \quad \widetilde{H} = \widetilde{H}^T, \quad \text{where} \quad t = \begin{bmatrix} t_{11} \\ 0 \end{bmatrix}, \quad u = \begin{bmatrix} t_{12} \\ t_{22} \end{bmatrix}.$$

We can take $\widetilde{H} := (\|u\|_2/\|t\|_2)P$, where $P \in \mathbb{R}^{2\times 2}$ is either a suitably chosen Householder matrix, or the identity if t is a positive multiple of u. Then

$$\begin{aligned} \|H\|_2 = \|\widetilde{H}\|_2 &= \|u\|_2/\|t\|_2 \\ &= \|r\|_2/\|y\|_2 = \|Gy\|_2/\|y\|_2 \le \|G\|_2, \end{aligned}$$

and $\|H\|_F = \|\widetilde{H}\|_F \le \sqrt{2}\|\widetilde{H}\|_2 \le \sqrt{2}\|G\|_2 \le \sqrt{2}\|G\|_F$.

(b) We can assume, without loss of generality, that

$$|y_1| \le |y_2| \le \cdots \le |y_n|. \tag{A.9}$$

Define the off-diagonal of H by $h_{ij} = h_{ji} = g_{ij}$ for $j > i$, and let $h_{11} = g_{11}$. The ith equation of the constraint $Hy = Gy$ will be satisfied if

$$h_{ii}y_i = g_{ii}y_i + \sum_{j=1}^{i-1}(g_{ij} - g_{ji})y_j. \tag{A.10}$$

If $y_i = 0$ set $h_{ii} = 0$; then (A.10) holds by (A.9). Otherwise, set

$$h_{ii} = g_{ii} + \sum_{j=1}^{i-1}(g_{ij} - g_{ji})\frac{y_j}{y_i}, \tag{A.11}$$

which yields

$$|h_{ii}| \le \epsilon |a_{ii}| + 2\epsilon \sum_{j=1}^{i-1} |a_{ij}| \le 3\epsilon |a_{ii}|,$$

by the diagonal dominance.

(c) Let $D = \mathrm{diag}(a_{ii}^{1/2})$ and $A =: D\widetilde{A}D$. Then $\widetilde{a}_{ii} \equiv 1$ and $|\widetilde{a}_{ij}| \le 1$ for $i \ne j$. The given condition can be written $(\widetilde{A} + \widetilde{G})\widetilde{y} = \widetilde{b}$, $|\widetilde{G}| \le \epsilon |\widetilde{A}|$, where $\widetilde{y} = Dy$ and $\widetilde{b} = D^{-1}b$. Defining $\widetilde{H}$ as in the proof of (b), we find from (A.11) that $|\widetilde{h}_{ii}| \le \epsilon + 2\epsilon(i-1) \le (2n-1)\epsilon$, so that $|\widetilde{H}| \le (2n-1)|\widetilde{A}|$. With $H := D\widetilde{H}D$, we find that $H = H^T$, $(A+H)y = b$, and $|H| \le (2n-1)\epsilon|A|$.

7.13. By examining the error analysis for an inner product (§3.1), it is easy to see that for any $x \in \mathbb{R}^n$,

$$|fl(Ax) - Ax| \le \mathrm{diag}(\gamma_{w_i})|A||x|.$$

Hence $\widehat{r} = fl(b - A\widehat{x})$ satisfies

$$\widehat{r} = r + \Delta r, \qquad |\Delta r| \le \Gamma(|A||\widehat{x}| + |b|), \tag{A.12}$$

which is used in place of (7.30) to obtain the desired bound.

7.14. (a) Writing $A^{-1} = (\alpha_{ij})$, we have

$$\Delta x_i = -\sum_{j=1}^n \alpha_{ij} \sum_{k=1}^n \epsilon_{jk} e_{jk} x_k + \sum_{j=1}^n \alpha_{ij} \delta_j f_j.$$

By the independence of the ϵ_{ij} and δ_i,

$$\mathcal{E}(\Delta x_i^2) = \sigma^2 \left(\sum_{j=1}^n \sum_{k=1}^n \alpha_{ij}^2 e_{jk}^2 x_k^2 + \sum_{j=1}^n \alpha_{ij}^2 f_j^2 \right) = \sigma^2 \left([A^{-1}][E][x] + [A^{-1}][f] \right)_i.$$

Hence

$$\mathcal{E}(\|\Delta x\|_2^2) = \sum_{i=1}^n \mathcal{E}(\Delta x_i^2) = \sigma^2 \|\, [A^{-1}][E][x] + [A^{-1}][f] \,\|_1.$$

(b) A traditional condition number would be based on the maximum of $\|\Delta x\|_2/(\sigma\|x\|_2)$ over all "perturbations to the data of relative size σ". The expression $\mathrm{condexp}(A,b)$ uses the "expected perturbation" $\sqrt{\mathcal{E}(\|\Delta x\|_2^2)}/(\sigma\|x\|_2)$ rather than the worst case.

(c) For $e_{ij} \equiv \|A\|_2$ and $f_i \equiv \|b\|_2$ we have

$$[A^{-1}][E][x] + [A^{-1}][f] = \left(\|A\|_2^2 \|x\|_2^2 + \|b\|_2^2\right)[A^{-1}] \begin{bmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{bmatrix},$$

so

$$\frac{\left(\|[A^{-1}][E][x] + [A^{-1}][f]\|_1\right)^{1/2}}{\|x\|_2} = \left(\|A\|_2^2 + \frac{\|b\|_2^2}{\|x\|_2^2} \right)^{1/2} \|A^{-1}\|_F.$$

But

$$\kappa_{A,b}(A,x) = \left(\|A\|_2 + \frac{\|b\|_2}{\|x\|_2} \right) \|A^{-1}\|_2.$$

Hence

$$2^{-1/2}\kappa_{A,b}(A,x) \le \mathrm{condexp}(A,x) \le \sqrt{n}\,\kappa_{A,b}(A,x).$$

This inequality shows that when perturbations are measured normwise there is little difference between the average and worst-case condition numbers.

7.15. We have

$$\begin{aligned}\|A \circ A^{-T}\|_2 &= \|D_1AD_2 \circ D_1^{-1}A^{-T}D_2^{-1}\|_2 = \|(D_1AD_2) \circ (D_1AD_2)^{-T}\|_2 \\ &\le \|D_1AD_2\|_2\|(D_1AD_2)^{-T}\|_2 = \kappa_2(D_1AD_2).\end{aligned}$$

Note that this result gives a lower bound for the optimal condition number, while Bauer's result in Problem 7.10(c) gives an upper bound. There is equality for diagonal A, trivially. For triangular A we have $\|A \circ A^{-T}\|_2 = 1 = \rho(|A||A^{-1}|)$ and so there is equality in view of Problem 7.10(e).

8.1. Straightforward.

8.2. Let

$$T(\lambda) = \begin{bmatrix} \lambda^{-1} & 1 & 1 \\ 0 & \lambda^{-1} & \lambda^{-1} \\ 0 & 0 & \lambda^{-2} \end{bmatrix}.$$

Then

$$T(\lambda)^{-1} = \begin{bmatrix} \lambda & -\lambda^2 & 0 \\ 0 & \lambda & -\lambda^2 \\ 0 & 0 & \lambda^2 \end{bmatrix}, \quad M(T(\lambda))^{-1} = \begin{bmatrix} \lambda & \lambda^2 & 2\lambda^3 \\ 0 & \lambda & \lambda^2 \\ 0 & 0 & \lambda^2 \end{bmatrix}.$$

As $\lambda \to \infty$, $\|M(T(\lambda))^{-1}\|/\|T(\lambda)^{-1}\| \sim \lambda$. This 3×3 example can be extended to an $n \times n$ one by padding with an identity matrix.

8.3. The bound follows from Theorem 8.10, since $\big(M(U)^{-1}\big)_{ij} \le 2^{j-i-1}$ $(j > i)$. Using $\|b\|_\infty \le \|U\|_\infty \|x\|_\infty \le n\|x\|_\infty$ we get a similar result to that given by Theorem 8.7.

8.4. Assume T is upper triangular, and write $T = D - U$, where $D = \mathrm{diag}(t_{ii}) \ge 0$ and $U \ge 0$ is strictly upper triangular. Then, using the fact that $(D^{-1}U)^n = 0$,

$$\begin{aligned}|T^{-1}||T| &= (I - D^{-1}U)^{-1}D^{-1} \cdot (D + U) \\ &= \sum_{i=0}^{n-1} (D^{-1}U)^i \cdot (I + D^{-1}U) \\ &= \sum_{i=0}^{n-1} (D^{-1}U)^i + \sum_{i=1}^{n-1} (D^{-1}U)^i.\end{aligned}$$

Now $0 \le b = Tx = (D - U)x$, so $Dx \ge Ux$, that is, $x \ge D^{-1}Ux$. Hence

$$|T^{-1}||T|x = \sum_{i=0}^{n-1} (D^{-1}U)^i x + \sum_{i=1}^{n-1} (D^{-1}U)^i x \le (n + n - 1)x,$$

which gives the result, since $x = T^{-1}b \ge 0$.

8.5. Use (8.4).

8.7. (a) Write

$$\|A^{-1}\|_\infty = \max_{x \ne 0} \frac{\|A^{-1}x\|_\infty}{\|x\|_\infty} =: \frac{\|b\|_\infty}{\|y\|_\infty},$$

where $Ab = y$. Let $|b_k| = \|b\|_\infty$. From

$$a_{kk}b_k = y_k - \sum_{j \ne k} a_{kj}b_j,$$

it follows that

$$|a_{kk}||b_k| \le \|y\|_\infty + \sum_{j\ne k} |a_{kj}|\|b\|_\infty,$$

and hence that $\|b\|_\infty/\|y\|_\infty \le 1/\alpha_k \le 1/(\min_i \alpha_i)$.

(b) Apply the result of (a) to AD and use the inequality $\|A^{-1}\|_\infty \le \|(AD)^{-1}\|_\infty \|D\|_\infty$.

(c) If A is triangular then we can achieve any diagonal dominance factors β_i we like, by appropriate column scaling. In particular, we can take $\beta_i \equiv 1$ in (b), which requires $M(A)d = e$, where $d = De$. Then $\|D\|_\infty = \|d\|_\infty = \|M(A)^{-1}e\|_\infty$, so the bound is $\|A^{-1}\|_\infty \le \|M(A)^{-1}e\|_\infty$, as required.

8.8. (a) Using the formula $\det(I+xy^T) = 1 + y^T x$ we have $\det(A+\alpha e_i e_j^T) = \det(A(I+\alpha A^{-1}e_i e_j^T)) = \det(A)(1+\alpha e_j^T A^{-1} e_i)$. Hence we take $\alpha_{ij} = -(e_j^T A^{-1} e_i)^{-1}$ if $e_j^T A^{-1} e_i \ne 0$; otherwise there is no α_{ij} that makes $A + \alpha_{ij} e_i e_j^T$ singular.

It follows that the "best" place to perturb A to make it singular (the place that gives the smallest α_{ij}) is in the (s,r) position, where the element of largest absolute value of A^{-1} is in the (r,s) position.

(b) The off-diagonal elements of T_n^{-1} are given by $(T_n^{-1})_{ij} = 2^{j-i-1}$. Hence, using part (a), $T_n + \alpha e_n e_1^T$ is singular, where $\alpha = -2^{2-n}$. In fact, T_n is also made singular by subtracting 2^{1-n} from all the elements in the first column.

8.9. Here is Zha's proof. If $s = 1$ the result is obvious, so assume $s < 1$. Define the n-vectors

$$v^T = \begin{bmatrix} 0, & \dots, & 0, & \frac{1}{\sqrt{2}}, -\frac{1}{\sqrt{2}} \end{bmatrix}$$

and

$$u^T = \begin{bmatrix} 0, & \dots, & 0, & \sqrt{\frac{1+c}{2}}, \frac{-s}{\sqrt{2(1+c)}} \end{bmatrix}$$

and let $\sigma = s^{n-2}\sqrt{1+c}$. It is easy to check that

$$U_n(\theta)v = \sigma u, \quad U_n(\theta)^T u = \sigma v,$$

which shows that σ is a singular value of $U_n(\theta)$. With σ_i denoting the ith largest singular value,

$$\sigma_n(U_n(\theta)) \le s^{n-1} < \sigma = s^{n-2}\sqrt{1+c}.$$

Now we prove by induction that $\sigma_{n-1}(U_n(\theta)) = \sigma$. For $n = 2$ it is easy to check by direct computation that $\sigma_1(U_2(\theta)) = \sqrt{1+c}$. Using the interlacing property of the singular values [509, 1996, §8.6.1] and the inductive assumption, we have

$$\sigma_{n-2}(U_n(\theta)) \ge \sigma_{n-2}(U_{n-1}(\theta)) = s^{n-3}\sqrt{1+c} > \sigma.$$

Therefore $\sigma_{n-1}(U_n(\theta)) = \sigma$.

8.10. For a solver of this form, it is not difficult to see that

$$|fl(f_i(T,b)) - f_i(T,b)| \le c_n u \overline{f}_i(M(T), |b|) + O(u^2),$$

where $\overline{f}_i$ denotes f_i with all its coefficients replaced by their absolute values, and where $\overline{f}_i(M(T),|b|)$ is a rational expression consisting entirely of nonnegative terms. This is the required bound expressed in a different notation. An example (admittedly, a contrived one) of a solver that does not satisfy (8.21) is, for a 2×2 lower triangular system $Lx = b$,

$$x_1 = b_1/l_{11}, \quad x_2 = \big((b_2 - b_1)(l_{11} + l_{21}) + b_1 l_{11} - b_2 l_{21}\big)/(l_{11} l_{22}).$$

9.1. The proof is by induction. Assume the result is true for matrices of order $n-1$, and suppose

$$\begin{bmatrix} \alpha & a^T \\ b & C \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ l & M \end{bmatrix} \begin{bmatrix} \alpha & a^T \\ 0 & V \end{bmatrix} \in \mathbb{R}^{n\times n}$$

is a unique LU factorization. Then $MV = C - la^T$ has a unique LU factorization, and so by the inductive hypothesis has nonsingular leading principal submatrices of order 1 to $n-2$. Thus $v_{ii} \neq 0$, $i = 1\colon n-2$. If $\alpha = 0$ then $b = 0$ and l is arbitrary subject to $C - la^T$ having an LU factorization. But $C - (l + \Delta l)a^T$ has nonsingular leading principal submatrices of order 1 to $n-2$ for sufficiently small Δl (irrespective of a), so has an LU factorization for sufficiently small Δl. Thus if $\alpha = 0$ we have a contradiction to the uniqueness of the factorization. Hence $\alpha \neq 0$, which completes the proof.

9.2. $A(\sigma)$ fails to have an LU factorization without pivoting only if one of its leading principal submatrices is singular, that is, if $\sigma \in \lambda\bigl(A(\sigma)(1\colon k, 1\colon k)\bigr)$, for $k \in \{1, 2, \dots, n-1\}$. There are thus $1 + 2 + \cdots + (n-1) = \frac{1}{2}(n-1)n$ "danger" values of σ, which may not all be distinct.

9.3. If $0 \notin F(A)$ then all the principal submatrices of A must be nonsingular, so the result follows by Theorem 9.1. Note that A may have a unique LU factorization even when 0 is in the field of values, as shown by the matrices

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \quad (z = e_2) \quad \text{and} \quad A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 0 & 0 \\ 1 & 0 & 1 \end{bmatrix} \quad (z = e_2),$$

so the implication is only one way. Note also that $0 \notin F(A)$ iff $e^{i\theta}A$ has positive definite Hermitian part for some real θ (Horn and Johnson [637, 1991, Thm. 1.3.5]).

9.4. The changes are minor. Denoting by $\widehat{P}$ and $\widehat{Q}$ the computed permutations, the result of Theorem 9.3 becomes

$$\widehat{L}\widehat{U} = \widehat{P}A\widehat{Q} + \Delta A, \qquad |\Delta A| \le \gamma_n |\widehat{L}||\widehat{U}|,$$

and that of Theorem 9.4 becomes

$$(A + \Delta A)\widehat{x} = b, \qquad |\Delta A| \le \gamma_{3n} \widehat{P}^T |\widehat{L}||\widehat{U}|\widehat{Q}^T.$$

9.5.

$$A = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} = LU; \qquad |L||U| = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}.$$

9.6. Since A is nonsingular and every submatrix has a nonnegative determinant, the determinantal inequality shows that $\det(A(1\colon p, 1\colon p)) > 0$ for $p = 1\colon n-1$, which guarantees the existence of an LU factorization (Theorem 9.1). That the elements of L and U are nonnegative follows from (9.2).

GE computes $a_{ij}^{(k+1)} = a_{ij}^{(k)} - m_{ik}a_{kj}^{(k)} = a_{ij}^{(k)} - l_{ik}u_{kj} \le a_{ij}^{(k)}$, since $l_{ik}, u_{kj} \ge 0$. Thus $a_{ij} = a_{ij}^{(1)} \ge a_{ij}^{(2)} \ge \cdots \ge a_{ij}^{(r)}$, $r = \min(i,j)$. For $i > j$, $a_{ij}^{(r)} \ge a_{ij}^{(r+1)} = \cdots = a_{ij}^{(n)} = 0$; for $j \ge i$, $a_{ij}^{(r)} = \cdots = a_{ij}^{(n)} = u_{ij} \ge 0$. Thus $0 \le a_{ij}^{(k)} \le a_{ij}$ for all i, j, k and hence $\rho_n \le 1$. But $\rho_n \ge 1$, so $\rho_n = 1$.

9.7. The number of square submatrices is

$$\begin{aligned}\sum_{k=1}^{n}\binom{n}{k}^2 &= \binom{2n}{n} \qquad ([517,\ 1994,\ \S5.1])\\ &= \frac{(2n)!}{(n!)^2}\\ &\sim \frac{\sqrt{2\pi(2n)}\,(2n/e)^{2n}}{\left(\sqrt{2\pi n}\,(n/e)^n\right)^2} \qquad \text{(Stirling's approximation)}\\ &\sim \frac{1}{\sqrt{n}}\left(\frac{2n/e}{n/e}\right)^{2n} \sim \frac{4^n}{\sqrt{n}} = O(4^n).\end{aligned}$$

The number of rectangular submatrices is

$$\sum_{k=1}^{n}\sum_{l=1}^{n}\binom{n}{k}\binom{n}{l} = \left(\sum_{k=1}^{n}\binom{n}{k}\right)^2 = (2^n-1)^2,$$

which, interestingly, is not many more than the number of square submatrices.

9.8. The given fact implies that JAJ is totally nonnegative. Hence it has an LU factorization $JAJ = LU$ with $L \geq 0$ and $U \geq 0$. This means that $A = (JLJ)(JUJ) \equiv \widetilde{L}\widetilde{U}$ is an LU factorization, and $|\widetilde{L}||\widetilde{U}| = LU = JAJ = |A|$.

9.9. With l_j the jth column of L and u_i^T the ith row of U, we have

$$A^{(k)} = A - l_1u_1^T - \cdots - l_{k-1}u_{k-1}^T = A - L(:,1\colon k-1)U(1\colon k-1,:),$$

hence

$$|A^{(k)}| \leq |A| + |L(:,1\colon k-1)||U(1\colon k-1,:)| \leq |A| + |L||U|.$$

Equating (i,j) elements gives

$$|a_{ij}^{(k)}| \leq |a_{ij}| + \|\,|L||U|\,\|_\infty,$$

from which the result follows on dividing by $\max_{i,j}|a_{ij}|$.

9.10. By inspecting the equations in Algorithm 9.2 we see that the computed LU factors satisfy

$$A - \widehat{L}\widehat{U} = \epsilon\,\widehat{l}_{ij}\widehat{u}_{jj}e_ie_j^T.$$

Since $\widehat{L}\widehat{U}\widehat{x} = b$, we have, writing $\alpha = \epsilon\,\widehat{l}_{ij}\widehat{u}_{jj}$ and using the Sherman–Morrison formula (Problem 26.2),

$$\widehat{x} = (A - \alpha e_ie_j^T)^{-1}b = \left(A^{-1} + \frac{\alpha A^{-1}e_ie_j^TA^{-1}}{1-\alpha e_j^TA^{-1}e_i}\right)b = x + \frac{\alpha A^{-1}e_ix_j}{1-\alpha e_j^TA^{-1}e_i}.$$

So

$$x - \widehat{x} = -\frac{\epsilon\,\widehat{l}_{ij}\widehat{u}_{jj}x_j}{1-\epsilon\,\widehat{l}_{ij}\widehat{u}_{jj}A^{-1}(j,i)}A^{-1}(:,i).$$

The error $x - \widehat{x}$ is a rational function of ϵ and is zero if $x_j = 0$, but it will typically be of order $\epsilon\,\kappa_\infty(A)\|x\|_\infty$.

9.11. $\alpha(B) = \alpha(A)$, and $B^{-1} = \frac{1}{2}\left[\begin{smallmatrix} A^{-1} & A^{-1}\\ A^{-1} & -A^{-1}\end{smallmatrix}\right]$, so $\beta(B) = \frac{1}{2}\beta(A)$. Hence $\theta(B) = (\alpha(B)\beta(B))^{-1} = 2\theta(A)$. Taking $A = S_n$, $g(2n) \geq \rho_{2n}^c(B) \geq \theta(B) = 2\theta(S_n) = n+1$.

9.12. First, the size or accuracy of the pivots is not the fundamental issue. The error analysis shows that instability corresponds to large elements in the intermediate matrices or in L or U. Second, in $PAQ = LU$, u_{nn}^{-1} is an element of A^{-1} (see (9.11)), so it is not necessarily true that the pivoting strategy can force u_{nn} to be small. Third, from a practical point of view it is good to obtain small final pivots because they reveal near singularity of the matrix.

9.13. Because the rows of U are the pivot rows, μ_j is the maximum number of times any element in the jth column of A was modified during the reduction. Since the multipliers are bounded by τ^{-1}, the bound for $\max_i |a_{ij}^{(k)}|$ follows easily. Thus

$$\rho_n \leq (1+\tau^{-1})^{\max_j \mu_j}.$$

10.1. Observe that if $x = \alpha e_i - e_j$, where e_i is the ith unit vector, then for any α we have, using the symmetry of A,

$$0 < x^T A x = \alpha^2 a_{ii} - 2\alpha a_{ij} + a_{jj}.$$

Since the right-hand side is a quadratic in α, the discriminant must be negative, that is, $4a_{ij}^2 - 4a_{ii}a_{jj} < 0$, which yields the desired result. This result can also be proved using the Cholesky decomposition and the Cauchy–Schwarz inequality: $A = R^T R \Rightarrow a_{ij} = r_i^T r_j \leq \|r_i\|_2 \|r_j\|_2 = (a_{ii}a_{jj})^{1/2}$ (there is strict inequality for $i \neq j$, since $r_i \neq \alpha r_j$). The inequality implies that $|a_{ij}| \leq \max(a_{ii}, a_{jj})$, which shows that $\max_{i,j} |a_{ij}| = \max_i a_{ii}$, that is, the largest element of A lies on the diagonal.

10.2. Compute the Cholesky factorization $A = R^T R$, solve $R^T y = x$, then compute $y^T y$. In addition to being more efficient than forming A^{-1} explicitly, this approach guarantees a nonnegative result.

10.3. Let $s = c - \sum_{i=1}^{k-1} a_i b_i$. By Lemma 8.4,

$$\widehat{s}(1+\theta_{k-1}^{(0)}) = c - \sum_{i=1}^{k-1} a_i b_i (1+\theta_{k-1}^{(i)}).$$

Then $\widehat{y} = fl(\sqrt{\widehat{s}}) = \sqrt{\widehat{s}}(1+\delta)$, $|\delta| \leq u$. Hence

$$\widehat{y}^2(1+\theta_{k+1}) = \frac{\widehat{y}^2}{(1+\delta)^2}(1+\theta_{k-1}^{(0)}) = c - \sum_{i=1}^{k-1} a_i b_i (1+\theta_{k-1}^{(i)}),$$

as required.

10.4. $A = \left[\begin{smallmatrix} \alpha & a^T \\ a & C \end{smallmatrix}\right] \Rightarrow A^{(2)} = \left[\begin{smallmatrix} \alpha & a^T \\ 0 & B \end{smallmatrix}\right]$, where $B = C - aa^T/\alpha$. Let $y = [1, \theta z]^T$. Then $0 < y^T A y = \theta^2 z^T C z + 2\theta z^T a + \alpha$ for all θ and so the discriminant $4(z^T a)^2 - 4\alpha z^T C z < 0$ if $z \neq 0$, that is, $z^T B z = z^T C z - \frac{1}{\alpha}(z^T a)^2 > 0$ (since $\alpha = a_{11} > 0$). This shows that B is positive definite.

By induction, the elimination succeeds (i.e., all pivots are nonzero) and all the reduced submatrices are positive definite. Hence $a_{rr}^{(r)} > 0$ for all r. Therefore

$$a_{kk}^{(r+1)} = a_{kk}^{(r)} - \frac{a_{kr}^{(r)}}{a_{rr}^{(r)}} a_{rk}^{(r)} = a_{kk}^{(r)} - \frac{{a_{rk}^{(r)}}^2}{a_{rr}^{(r)}} \leq a_{kk}^{(r)}.$$

Thus $a_{kk} = a_{kk}^{(1)} \geq \cdots \geq a_{kk}^{(k)} > 0$. Since the largest element of a positive definite matrix lies on the diagonal (Problem 10.1), for any i, j, k there exists r such that

$$|a_{ij}^{(k)}| \leq a_{rr}^{(k)} \leq \cdots \leq a_{rr}^{(1)} = a_{rr} \leq \max_{i,j} |a_{ij}|,$$

which shows that the growth factor $\rho_n = 1$ (since $\rho_n \geq 1$).

10.5. From (10.8),

$$|\widehat{R}^T||\widehat{R}| \le (1-\gamma_{n+1})^{-1}(a_{ii}^{1/2}a_{jj}^{1/2}) \le (1-\gamma_{n+1})^{-1}(\max_i a_{ii}),$$

so

$$\|\,|\widehat{R}^T||\widehat{R}|\,\|_M \le (1-\gamma_{n+1})^{-1}\|A\|_M,$$

as required.

10.6. $W = A_{11}^{-1}A_{12} = R_{11}^{-1}R_{12}$, so $RZ = 0$ and hence $AZ = 0$. Z is of dimension $n \times (n-r)$ and of full rank, so it spans null(A).

10.7. The inequalities (10.13) follow from the easily verified fact that, for $j \ge k$,

$$a_{jj}^{(k)} = \sum_{i=k}^{\min(j,r)} r_{ij}^2.$$

10.8. Examples of indefinite matrices with nonnegative leading principal minors are

$$\begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}.$$

A necessary and sufficient condition for definiteness is that all the *principal minors* of A (of which there are 2^{n-1}) be nonnegative (not just the leading principal minors); see, e.g., Horn and Johnson [636, 1985, p. 405] or Mirsky [857, 1961, p. 405] (same page number in both books!).

10.9. For the matrix $Z = [-(R_{11}^{-1}R_{12})^T, I]^T \in \mathbb{R}^{n\times(n-k)}$ we have, from (10.15), $Z^TAZ = S_k(A)$ and so we can take $p = Ze_1$, the first column of Z.

10.10. Theorem 10.3 is applicable only if Cholesky succeeds, which can be guaranteed only if the (suitably scaled) matrix A is not too ill conditioned (Theorem 10.7). Therefore the standard analysis is *not* applicable to positive semidefinite matrices that are very close to being singular. Theorem 10.14 provides a bound on the residual after rank(A) stages and, in particular, on the computed Schur complement, which would be zero in exact arithmetic. The condition of Theorem 10.7 ensures that all the computed Schur complements are positive definite, so that even if magnification of errors by a factor $\|W\|_2^2$ occurs, it is absorbed by the next part of the Cholesky factor. The proposed appeal to continuity is simply not valid.

10.11. For any nonzero $x \in \mathbb{C}^n$ we have

$$0 < \mathrm{Re}\,[\,x_1^* \quad x_2^*\,]\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Putting $x_1 = -A_{11}^{-1}A_{12}x_2$ we obtain

$$0 < \mathrm{Re}\,[\,x_1^* \quad x_2^*\,]\begin{bmatrix} 0 \\ -A_{21}^*A_{11}^{-1}A_{12}x_2 + A_{22}x_2 \end{bmatrix} = \mathrm{Re}\,x_2^*(A_{22} - A_{21}^*A_{11}^{-1}A_{12})x_2,$$

which shows that S is positive definite.

11.1. If a nonsingular pivot matrix E of dimension 1 or 2 cannot be found, then all 1×1 and 2×2 principal submatrices of the symmetric matrix A are singular, and this is easily seen to imply that A is the zero matrix.

11.2. The analysis in §11.1 shows that for a 2×2 pivot E, $\det(E) \le (\alpha^2 - 1)\mu_0^2$ for complete pivoting and $\det(E) \le (\alpha^2 - 1)\omega_r^2$ for partial pivoting ($r = 1$) and rook pivoting. Now $\alpha^2 - 1 < 0$ and μ_0 and ω_r are nonzero if a 2×2 pivot is needed. Hence $\det(E) < 0$, which means that E has one positive and one negative eigenvalue. Note that if A is positive definite it follows that all pivots are 1×1.

If the block diagonal factor has p_+ positive 1×1 diagonal blocks, p_- negative 1×1 diagonal blocks, p_0 zero 1×1 diagonal blocks, and q 2×2 diagonal blocks, then the inertia is $(+, -, 0) = (p_+ + q, p_- + q, p_0)$.

Denote a 2×2 pivot by

$$E = \begin{bmatrix} a & b \\ b & c \end{bmatrix},$$

and consider partial pivoting. We know $\det(E) = ac - b^2 < 0$ and $|b| \ge |a|$, so the formula $\det(E) = [(a/b)c - b]b$ minimizes the risk of overflow. Similarly, the formula

$$E^{-1} = \frac{1}{b[(a/b)(c/b) - 1]} \begin{bmatrix} c/b & -1 \\ -1 & a/b \end{bmatrix}$$

helps to avoid overflow; this is the formula used in LINPACK's `xSIDI` and LAPACK's `xSYTRI`. The same formulae are suitable for rook pivoting and complete pivoting because then $|b| \ge \max(\,|a|, |c|\,)$.

11.3. The partial pivoting strategy simplifies as follows: if $|a_{11}| \ge \alpha|a_{21}|$ use a 1×1 pivot a_{11}, if $|a_{22}| \ge \alpha|a_{12}|$ use a 1×1 pivot a_{22}, else use a 2×2 pivot, that is, do nothing.

11.4. There may be interchanges, because the tests $|a_{11}| \ge \alpha\omega_1$ and $\alpha\omega_1^2 \le |a_{11}|\omega_r$ can both fail, for example for $A = \left[\begin{smallmatrix} 1 & \theta \\ \theta & 2\theta^2 \end{smallmatrix}\right]$ with $\theta > \alpha^{-1}$. But there can be no 2×2 pivots, as they would be indefinite (Problem 11.2). Therefore the factorization is $PAP^T = LDL^T$ for a diagonal D with positive diagonal entries.

11.7. That the growth factor bound is unchanged is straightforward to check. No 2×2 pivots are used for a positive definite matrix because, as before (Problem 11.2), any 2×2 pivot is indefinite. To show that no interchanges are required for a positive definite matrix we show that the second test, $\alpha\omega_1^2 \le |a_{11}|\omega_r$, is always passed. The submatrix $\left[\begin{smallmatrix} a_{11} & a_{r1} \\ a_{r1} & a_{rr} \end{smallmatrix}\right]$ is positive definite, so $a_{11}a_{rr} - a_{r1}^2 > 0$. Hence

$$|a_{11}|\omega_r \ge a_{11}a_{rr} > a_{r1}^2 = \omega_1^2 \ge \alpha\omega_1^2,$$

as required.

11.8. Complete pivoting and rook pivoting both yield

$$PAP^T = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 0 & \epsilon \\ 0 & \epsilon & 1 \end{bmatrix} = \begin{bmatrix} 1 & & \\ 1 & 1 & \\ & -\epsilon & 1 \end{bmatrix} \begin{bmatrix} 1 & & \\ & -1 & \\ & & 1+\epsilon^2 \end{bmatrix} \begin{bmatrix} 1 & 1 & \\ & 1 & -\epsilon \\ & & 1 \end{bmatrix}.$$

Unlike for partial pivoting, $\|L\|_\infty$ is bounded independently of ϵ as $\epsilon \to 0$.

11.9. (a) The nonsingularity of A follows from the factorization

$$\begin{bmatrix} H & B^T \\ B & -G \end{bmatrix} = \begin{bmatrix} I & 0 \\ BH^{-1} & I \end{bmatrix} \begin{bmatrix} H & B^T \\ 0 & -(G + BH^{-1}B^T) \end{bmatrix},$$

since $G + BH^{-1}B^T$ is symmetric positive definite.

(b) It suffices to show that the first $n+m-1$ leading principal submatrices of $\Pi^T A \Pi$ are nonsingular for any permutation matrix Π. But these submatrices are of the form $\overline{\Pi}^T A_p \overline{\Pi}$, where A_p is a principal submatrix of A and $\overline{\Pi}$ is a permutation matrix. Any such A_p is of the form

$$A_p = \begin{bmatrix} H_p & B_p^T \\ B_p & -G_p \end{bmatrix},$$

where H_p and G_p are principal submatrices of H and G, respectively, and so are positive definite. Thus A_p is symmetric quasidefinite and hence nonsingular by (a), as required.

(c)

$$AS = \begin{bmatrix} H & -B^T \\ B & G \end{bmatrix},$$

so $(AS + (AS)^T)/2 = \mathrm{diag}(H, G)$, which is symmetric positive definite.

12.1.

$$|A||x| \le \|A\|_\infty \|x\|_\infty e \le \|A\|_\infty \|x\|_\infty \frac{|x|}{\min_i |x_i|} = \|A\|_\infty \frac{\max_i |x_i|}{\min_i |x_i|} |x|.$$

12.2. The inequality (12.5) yields, with $\widehat{x}_0 := 0$, and dropping the subscripts on the g_i and G_i,

$$|x - \widehat{x}_2| \le G|x - \widehat{x}_1| + g \le G^2|x - \widehat{x}_0| + Gg + g = G^2|x| + (G + I)g.$$

Now $G \approx |F| \le 2\gamma_n |A^{-1}||\widehat{L}||\widehat{U}| \lesssim 2\gamma_n |A^{-1}||B|A|$, where $B = |\widehat{L}||\widehat{L}^{-1}|$ can be assumed to have a modest ∞-norm. Now, using Problem 12.1,

$$\begin{aligned} G^2|x| &\lesssim 2n^2u^2|A^{-1}|(B|A||A^{-1}|B)|A||x| \\ &\lesssim 2n^2u^2 \|\, B|A||A^{-1}|B\,\|_\infty \sigma(A, x)|A^{-1}||A||x| \\ &\le 2n^2 \|B\|_\infty^2 \cdot u\,\mathrm{cond}(A^{-1})\sigma(A, x) \cdot u|A^{-1}||A||x| \\ &\lesssim 2n^2 \|B\|_\infty^2 \cdot u|A^{-1}||A||x|, \end{aligned}$$

under the conditions of Theorem 12.4. The term Gg can be bounded in a similar way. The required bound for $\|x - \widehat{x}_2\|_\infty / \|x\|_\infty$ follows.

13.1. The equations for the blocks of L and U are $U_{11} = A_{11}$ and

$$\left.\begin{aligned} L_{i,i-1} &= A_{i,i-1}U_{i-1,i-1}^{-1} \\ U_{ii} &= A_{ii} - L_{i,i-1}A_{i-1,i} \end{aligned}\right\} \quad i \ge 2.$$

First, consider the case where A is block diagonally dominant by columns. We prove by induction that

$$\|A_{k,k-1}\|\,\|U_{k-1,k-1}^{-1}\| \le 1, \quad k \ge 2.$$

which implies both the required bounds. This inequality clearly holds for $k = 2$; suppose it holds for $k = i$. We have

$$U_{ii} = A_{ii}(I - A_{ii}^{-1}L_{i,i-1}A_{i-1,i}) =: A_{ii}(I - P),$$

where

$$\begin{aligned} \|P\| &\le \|A_{ii}^{-1}\|\big(\|A_{i,i-1}\|\,\|U_{i-1,i-1}^{-1}\|\big)\|A_{i-1,i}\| \\ &\le \|A_{ii}^{-1}\|\,\|A_{i-1,i}\| \\ &\le 1 - \|A_{ii}^{-1}\|\,\|A_{i+1,i}\|, \end{aligned}$$

using the block diagonal dominance for the last inequality. Hence

$$\|U_{ii}^{-1}\| \leq \frac{\|A_{ii}^{-1}\|}{1-\|P\|} \leq \frac{1}{\|A_{i+1,i}\|},$$

as required.

The proof for A block diagonally dominant by rows is similar. The inductive hypothesis is that

$$\|U_{k-1,k-1}^{-1}\| \, \|A_{k-1,k}\| \leq 1,$$

and with P defined as before, we have

$$\begin{aligned}\|P\| &\leq \|A_{ii}^{-1}\| \, \|A_{i,i-1}\| \, \|U_{i-1,i-1}^{-1}\| \, \|A_{i-1,i}\| \\ &\leq \|A_{ii}^{-1}\| \, \|A_{i,i-1}\| \\ &\leq 1 - \|A_{ii}^{-1}\| \, \|A_{i,i+1}\|,\end{aligned}$$

giving $\|U_{ii}^{-1}\| \leq \|A_{i,i+1}\|^{-1}$, as required.

For block diagonal dominance by columns in the ∞-norm we have $\|L\|_\infty \leq 2$ and $\|U\|_\infty \leq 3\|A\|_\infty$, so block LU factorization is stable. If A is block diagonally dominant by rows, stability is assured if $\|A_{i,i-1}\|/\|A_{i-1,i}\|$ is suitably bounded for all i.

13.2. The block 2×2 matrix

$$\left[\begin{array}{cc|cc} \epsilon & 2 & 1 & 0 \\ 2 & \epsilon & 0 & 1 \\ \hline 1 & 0 & \epsilon & 2 \\ 0 & 1 & 2 & \epsilon \end{array}\right]$$

is block diagonally dominant by rows and columns in the 1- and ∞-norms for $\epsilon = 1/2$, but is not point diagonally dominant by rows or columns. The block 2×2 matrix

$$\left[\begin{array}{cc|cc} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 0 \\ \hline 1 & 0 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

is point diagonally dominant by rows and columns but not block diagonally dominant by rows or columns in the ∞-norm or 1-norm.

13.3. No. A counterexample is the first matrix in the solution to Problem 13.2, with $\epsilon = 1/2$, which is clearly not positive definite because the largest element does not lie on the diagonal.

13.4. From (13.2) it can be seen that $(A^{-1})_{21} = -S^{-1}A_{21}A_{11}^{-1}$, where the Schur complement $S = A_{22} - A_{21}A_{11}^{-1}A_{12}$. Hence

$$\|A_{21}A_{11}^{-1}\| \leq n\|S\| \, \|(A^{-1})_{21}\| \leq n\|S\| \, \|A^{-1}\|,$$

where the factor n comes from the fact that this norm is not consistent—see §6.2. S is the trailing submatrix that would be obtained after $r-1$ steps of GE. It follows immediately that $\|S\| \leq \rho_n\|A\|$.

For the last part, note that $\|S^{-1}\| \leq \|A^{-1}\|$, because S^{-1} is the $(2,2)$ block of A^{-1}, as is easily seen from (13.2).

13.5. The proof is similar to that of Problem 8.7(a). We will show that $\|A_{11}^{-T}A_{21}^T\|_\infty \le 1$. Let $y = A_{11}^{-T}A_{21}^T x$ and let $|y_k| = \|y\|_\infty$. The kth equation of $A_{11}^T y = A_{21}^T x$ gives

$$a_{kk}y_k = \sum_{i=r+1}^{n} a_{ik}x_{i-r} - \sum_{\substack{i=1\\ i\ne k}}^{r} a_{ik}y_i.$$

Hence

$$\|y\|_\infty = |y_k| \le \|x\|_\infty \sum_{i=r+1}^{n} \left|\frac{a_{ik}}{a_{kk}}\right| + \|y\|_\infty \sum_{\substack{i=1\\ i\ne k}}^{r} \left|\frac{a_{ik}}{a_{kk}}\right|,$$

which yields $\|y\|_\infty \le \|x\|_\infty$ in view of the column diagonal dominance, as required.

13.7. We have the block LU factorization

$$X = \begin{bmatrix} A & B \\ C & D \end{bmatrix} = \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ 0 & D - CA^{-1}B \end{bmatrix},$$

so that

$$\det(X) = \det(A)\det(D - CA^{-1}B).$$

Hence $\det(X) = \det(AD - ACA^{-1}B)$, which equals $\det(AD - CB)$ if C commutes with A.

13.8. The result is obtained by inverting

$$\begin{bmatrix} A & B \\ C & D \end{bmatrix}^{-1} = \begin{bmatrix} I & 0 \\ CA^{-1} & I \end{bmatrix} \begin{bmatrix} A & B \\ 0 & S \end{bmatrix}.$$

13.9. We have

$$\begin{bmatrix} I & A \\ B & I \end{bmatrix} = \begin{bmatrix} I & 0 \\ B & I \end{bmatrix} \begin{bmatrix} I & A \\ 0 & I - BA \end{bmatrix} = \begin{bmatrix} I & A \\ 0 & I \end{bmatrix} \begin{bmatrix} I - AB & 0 \\ B & I \end{bmatrix}.$$

Inverting both factorizations and equating (1,1) blocks gives the required relation. The Sherman–Morrison–Woodbury formula is obtained by setting $A = T^{-1}U$ and $B = W^{-1}V^T$.

14.2. The new bounds have norms in place of absolute values and the constants are different.

14.3. Immediate from $AX - I = A(XA - I)A^{-1}$.

14.4. With $0 < \epsilon \ll 1$, let

$$A = \begin{bmatrix} \dfrac{1}{\epsilon} & 1 \\ \dfrac{1}{\epsilon^2} - 1 & \dfrac{1}{\epsilon} \end{bmatrix}, \qquad X = \begin{bmatrix} 1 - \epsilon + \dfrac{2}{\epsilon} & -2 - \epsilon \\ 2 - \epsilon + \dfrac{1}{\epsilon} - \dfrac{1}{\epsilon^2} & -1 - \epsilon + \dfrac{1}{\epsilon} \end{bmatrix}.$$

Then

$$AX = \begin{bmatrix} O(\epsilon^{-2}) & O(\epsilon^{-1}) \\ O(\epsilon^{-3}) & O(\epsilon^{-2}) \end{bmatrix}$$

while

$$XA = \begin{bmatrix} 1 + \epsilon & -\epsilon \\ \epsilon & 1 - \epsilon \end{bmatrix}.$$

Hence $\|AX - I\|/\|XA - I\| \to \infty$ as $\epsilon \to 0$. Note that in this example *every* element of $AX - I$ is large.

14.5. We have $\widehat{x} = fl(\widehat{X}b) = \widehat{X}b + f_1$, where $|f_1| \le \gamma_n|\widehat{X}||b|$. Hence $A\widehat{x} = A\widehat{X}b + Af_1 = b + r_1$, where $|r_1| \le u|A||\widehat{X}||b| + \gamma_n|A||\widehat{X}||b| \le \gamma_{n+1}|A||\widehat{X}||b|$. Similarly,

$$\widehat{y} = fl(\widehat{Y}b) = \widehat{Y}b + f_2, \qquad |f_2| \le \gamma_n|\widehat{Y}||b|. \tag{A.13}$$

Hence $A\widehat{y} = A\widehat{Y}b + Af_2 = A(\widehat{Y}A)x + Af_2 = b + r_2$, where

$$|r_2| \le u|A||\widehat{Y}||A||x| + \gamma_n|A||\widehat{Y}||b| \le \gamma_{n+1}|A||\widehat{Y}||A||x|.$$

Clearly,

$$|x - \widehat{x}| \le (n+1)u|A^{-1}||A||A^{-1}||b| + O(u^2).$$

From (A.13), $\widehat{y} = \widehat{Y}Ax + f_2$, so

$$|y - \widehat{y}| \le u|\widehat{Y}||A||x| + \gamma_n|\widehat{Y}||b| \le (n+1)u|A^{-1}||A||x| + O(u^2).$$

The first conclusion is that the approximate left inverse yields the smaller residual bound, while the approximate right inverse yields the smaller forward error bound. Therefore which inverse is "better" depends on whether a small backward error or a small forward error is desired. The second conclusion is that neither approximate inverse yields a componentwise backward stable algorithm, despite the favourable assumptions on $\widehat{X}$ and $\widehat{Y}$. Multiplying by an explicit inverse is simply not a good way to solve a linear system.

14.6. Here is a hint: notice that the matrix on the front cover of the *LAPACK Users' Guide* has the form

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & -1 \\ 1 & -1 & 1 \end{bmatrix} \otimes \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \mathrm{diag}(L, A, P, A, C, K).$$

14.7. If the ith row of A contains all 1s then simply sum the elements in the ith row of the equation $AA^{-1} = I$.

14.8. $(A + iB)(P + iQ) = I$ is equivalent to $AP - BQ = I$ and $AQ + BP = 0$, or

$$\begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} P \\ Q \end{bmatrix} = \begin{bmatrix} I \\ 0 \end{bmatrix},$$

so X^{-1} is obtainable from the first n columns of Y^{-1}. The definiteness result follows from

$$\begin{aligned} (x+iy)^*(A+iB)(x+iy) &= x^T(Ax - By) + y^T(Ay + Bx) \\ &\quad + i[x^T(Ay + Bx) - y^T(Ax - By)] \\ &= [\,x^T \quad y^T\,] \begin{bmatrix} A & -B \\ B & A \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}, \end{aligned}$$

where we have used the fact that $A = A^T$ and $B = -B^T$. Doubling the dimension (from X to Y) multiplies the number of flops by a factor of 8, since the flop count for inversion is cubic in the dimension. Yet complex operations should, in theory, cost between about two and eight times as much as real ones (the extremes being for addition and division). The actual relative costs of inverting X and Y depend on the machine and the compiler, so it is not possible to draw any firm conclusions. Note also that Y requires twice the storage of X.

14.10. As in the solution to Problem 8.8, we have

$$\det(A + \alpha e_i e_j^T) = \det(A)(1 + \alpha e_j^T A^{-1} e_i).$$

If $(A^{-1})_{ji} = 0$, this expression is independent of α, and hence $\det(A)$ is independent of a_{ij}. (This result is clearly correct for a triangular matrix.) That $\det(A)$ can be independent of a_{ij} shows that $\det(A)$, or even a scaling of it, is an arbitrarily poor measure of conditioning, for $A + \alpha e_i e_j^T$ approaches a multiple of a rank-1 matrix as $\alpha \to \infty$.

14.11. That Hadamard's inequality can be deduced from QR factorization was noted by Householder [643, 1958, p. 341]. From $A = QR$ we have $q_k^T A = e_k^T R$, so that

$$|r_{kk}| = |q_k^T a_k| \le \|a_k\|_2.$$

Hadamard's inequality follows since $|\det(A)| = |\det(R)| = |r_{11} \dots r_{nn}|$. There is equality when $|q_k^T a_k| = \|a_k\|_2$ for $k = 1\colon n$, that is, when $a_k = \alpha_k q_k$, $k = 1\colon n$, for some scalars α_k. In other words, there is equality when R in the QR factorization is diagonal.

14.12. (a) Straightforward. (b) For $A = U(1)$, $\psi(A) = \sqrt{n!}$. For the Pei matrix, $\psi(A) = (\alpha^2 + n - 1)^{n/2} / \big((n + \alpha - 1)(\alpha - 1)^{n-1}\big)$.

14.13. (a) The geometric mean of $\sigma_1^2/2$, $\sigma_1^2/2$, σ_2^2, $\dots$, σ_{n-1}^2 is

$$\left(\frac{1}{4}\sigma_1^4\sigma_2^2 \dots \sigma_{n-1}^2\right)^{1/n} = \left(\frac{1}{4}\frac{\sigma_1^2}{\sigma_n^2}\sigma_1^2\sigma_2^2 \dots \sigma_n^2\right)^{1/n} = \left(\frac{1}{2}\kappa_2(A)|\det(A)|\right)^{2/n}.$$

Since the geometric mean does not exceed the arithmetic mean,

$$\left(\frac{1}{2}\kappa_2(A)|\det(A)|\right)^{2/n} \le \frac{1}{n}(\sigma_1^2 + \dots + \sigma_{n-1}^2) < \frac{1}{n}(\sigma_1^2 + \dots + \sigma_n^2) = \frac{1}{n}\|A\|_F^2,$$

which gives the required bound. (b) is trivial.

14.14. The key observation is that for a triangular system $Tx = b$, $T \in \mathbb{R}^{n \times n}$, the computed solution from substitution satisfies

$$(T + \Delta T)\widehat{x} = b + \Delta b, \qquad |\Delta T| \le \gamma_n |T|, \quad \operatorname{diag}(T) = 0, \quad |\Delta b| \le \gamma_n |b|. \tag{A.14}$$

This result holds for any order of evaluation and is proved (for any particular order of evaluation) using a variation on the proof of Lemma 8.2 in which we do not divide through by the product of $1 + \delta_i$ terms. Using (A.14) we have

$$\widehat{d} = (1 + \theta_n)\det(T)\big(\eta - (h + \Delta h)^T (T + \Delta T)^{-1}(y + \Delta y)\big),$$

where $|\theta_n| \le \gamma_n$, $|\Delta y| \le \gamma_{n-1}|y|$, and $|\Delta h| \le \gamma_{n-1}|h|$. But, since $\operatorname{diag}(\Delta T) = 0$, $\det(T) = \det(T + \Delta T)$, hence

$$\widehat{d} = \det(T + \Delta T)\big(\eta(1 + \theta_n) - (h + \widetilde{\Delta h})^T (T + \Delta T)^{-1}(y + \Delta y)\big),$$

where $|\widetilde{\Delta h}| \le \gamma_{2n-1}|h|$. The conclusion is that $\widehat{d} = \det(H + \Delta H)$, where $|\Delta H| \le \gamma_{2n-1}|H|$.

If $\widehat{f} = fl(\det(D^{-1}HD))$ then

$$\widehat{f} = \det(D^{-1}HD + \Delta), \quad |\Delta| \le \gamma_{2n-1}|D^{-1}HD| = \gamma_{2n-1}|D^{-1}||H|D|.$$

Thus

$$\widehat{f} = \det(D^{-1}(H + D\Delta D^{-1})D) = \det(H + D\Delta D^{-1}), \quad |D\Delta D^{-1}| \le \gamma_{2n-1}|H|.$$

A diagonal similarity therefore has no effect on the error bound and so there is no point in scaling H before applying Hyman's method.

14.15. We use the fact that $\det(A) = \prod_{i=1}^n \sigma_i(A)$, where $\sigma_i(A)$ is the ith largest singular value of A. We have

$$\frac{\det(A+\Delta A) - \det(A)}{\det(A)} = \prod_{i=1}^{n} \frac{\sigma_i(A+\Delta A)}{\sigma_i(A)} - 1.$$

Now, using standard singular value inequalities [509, 1996, §8.6],

$$\frac{\sigma_i(A+\Delta A)}{\sigma_i(A)} \le \frac{\sigma_i(A)+\sigma_1(\Delta A)}{\sigma_i(A)} \le 1 + \frac{\sigma_1(\Delta A)}{\sigma_n(A)} = 1 + \kappa_2(A)\frac{\|\Delta A\|_2}{\|A\|_2}$$

and

$$\frac{\sigma_i(A+\Delta A)}{\sigma_i(A)} \ge \frac{\sigma_i(A)-\sigma_1(\Delta A)}{\sigma_i(A)} \ge 1 - \frac{\sigma_1(\Delta A)}{\sigma_n(A)} = 1 - \kappa_2(A)\frac{\|\Delta A\|_2}{\|A\|_2}.$$

Hence

$$\prod_{i=1}^{n} \frac{\sigma_i(A+\Delta A)}{\sigma_i(A)} - 1 = \prod_{i=1}^{n}(1+\theta_i) - 1, \qquad |\theta_i| \le \kappa_2(A)\frac{\|\Delta A\|_2}{\|A\|_2}, \quad i = 1\colon n,$$

and the result follows from Lemma 3.1.

15.1. We have

$$\begin{aligned}
\|\,|A||A^{-1}||A|\,\|_\infty &= \|\,|A||A^{-1}||A|e\,\|_\infty \\
&= \|\,|A||A^{-1}|D_1 e\,\|_\infty, \qquad D_1 = \mathrm{diag}(|A|e) \\
&= \|\,|A||A^{-1}|D_1\,\|_\infty \\
&\approx \|\,|A||A^{-1}|D_1\,\|_1 \\
&= \|\,D_1|A^{-T}||A^T|\,\|_\infty \\
&= \|\,D_1|A^{-T}|D_2\,\|_\infty, \qquad D_2 = \mathrm{diag}(|A^T|e) \\
&= \|D_1 A^{-T} D_2\|_\infty,
\end{aligned}$$

where $\approx$ means "within a factor n of".

15.4. Straightforward, since L is unit lower triangular with $|l_{ij}| \le 1$.

15.7. Let $D = \mathrm{diag}(d_j)$, where $d_1 = 1$ and

$$d_j = \prod_{k=1}^{j-1} \frac{a_{k,k+1}}{a_{k+1,k}}, \qquad j = 2\colon n.$$

Then $T = DA$ is tridiagonal, symmetric and irreducible. By applying Theorem 15.9 and using symmetry, we find that

$$(A^{-1})_{ij} = \begin{cases} x_i y_j d_j, & i \le j, \\ y_i x_j d_j, & i \ge j. \end{cases}$$

There is one degree of freedom in the vectors x and y, which can be expended by setting $x_1 = 1$, say.

16.1. The equation would imply, on taking traces, $0 = \mathrm{trace}(I)$, which is false.

16.2. It is easily checked that the differential equation given in the hint has the solution $Z(t) = e^{At}Ce^{Bt}$. Integrating the differential equation between 0 and ∞ gives, assuming that $\lim_{t\to\infty} Z(t) = 0$,

$$-C = A\left(\int_0^\infty Z(t)\right) dt + \left(\int_0^\infty Z(t)dt\right) B.$$

Hence $-\int_0^\infty Z(t)\,dt = -\int_0^\infty e^{At}Ce^{Bt}\,dt$ satisfies the Sylvester equation. For the Lyapunov equation, the integral $-\int_0^\infty e^{At}(-C)e^{A^Tt}\,dt$ exists under the assumption on A, and the corresponding quadratic form is easily seen to be positive.

16.3. Since

$$\|AX + XA^T\|_F = \|(AX + XA^T)^T\|_F = \|AX^T + X^TA^T\|_F,$$

X^T is a minimizer if X is. If $X + X^T = 0$ then X is a skew-symmetric minimizer. Otherwise, note that the minimization problem is equivalent to finding the right singular vector corresponding to the smallest singular value $\sigma_{\min}$ of $P = I \otimes A + A \otimes I$. Since $\mathrm{vec}(X)$ and $\mathrm{vec}(X^T)$ (suitably normalized) are both singular vectors corresponding to $\sigma_{\min}$, so is their sum, $\mathrm{vec}(X + X^T) \neq 0$. Thus the symmetric matrix $X + X^T$ is a minimizer.

Byers and Nash [195, 1987] investigate conditions under which a symmetric minimizer exists.

16.4. `xLASY2` uses Gaussian elimination with complete pivoting to solve the 2×2 or 4×4 linear system that arises on converting to the Kronecker product form (16.2). For complete pivoting on a 4×4 matrix the growth factor is bounded by 4 (§9.4), versus 8 for partial pivoting. The reasons for using complete pivoting rather than partial pivoting are that, first, the small increase in cost is regarded as worthwhile for a reduction in the error bound by a factor of 2 and, second, complete pivoting reveals the rank better than partial pivoting, enabling better handling of ill-conditioned systems [46, 1993, p. 78].

17.1. Since $\rho(B) < 1$, a standard result (Problem 6.8) guarantees the existence of a consistent norm $\|\cdot\|_\rho$ for which $\|B\|_\rho < 1$. The series $\sum_{k=0}^\infty \|B^k\|_\rho \le \sum_{k=0}^\infty \|B\|_\rho^k = (1 - \|B\|_\rho)^{-1}$ is clearly convergent, and so, by the equivalence of norms, $\sum_{k=0}^\infty \|B^k\|$ is convergent for any norm.

Since $(|B^k|)_{ij} \le \|B^k\|_\infty$, the convergence of $\sum_{k=0}^\infty \|B^k\|_\infty$ ensures that of $\sum_{k=0}^\infty |B^k|$. (The convergence of $\sum_{k=0}^\infty |B^k|$ can also be proved directly using the Jordan canonical form.)

18.1. Let X_n be the $n \times n$ version of the upper triangular matrix

$$\begin{bmatrix} 1 & -\theta & -\theta & -\theta \\ & 1 & -\theta & -\theta \\ & & 1 & -\theta \\ & & & 1 \end{bmatrix} D^{-1}, \qquad \theta > 0,$$

where $D = \mathrm{diag}(\sqrt{1 + (j-1)\theta^2})$, so that $\|X_n(:,j)\|_2 = 1$, $j = 1\colon n$. From (8.4), X_n^{-1} is the obvious $n \times n$ analogue of

$$D \begin{bmatrix} 1 & \theta & \theta(\theta+1) & \theta(\theta+1)^2 \\ & 1 & \theta & \theta(\theta+1) \\ & & 1 & \theta \\ & & & 1 \end{bmatrix}.$$

Let $\Lambda = \mathrm{diag}(0, \dots, 0, \lambda)$, with $|\lambda| < 1$. Then

$$\begin{aligned}\|A^k\|_2 &= \|X_n \Lambda^k X_n^{-1}\|_2 = |\lambda|^k \|X_n e_n e_n^T X_n^{-1}\|_2 \\ &= |\lambda|^k \|X_n(:, n)\|_2 \|X_n^{-1}(n, :)\|_2 = |\lambda|^k \sqrt{1 + (n-1)\theta^2}.\end{aligned}$$

But

$$\kappa_2(X_n)\rho(A)^k \geq |\lambda|^k \max_{i,j} |x_{ij}| \max_{i,j} |(X_n^{-1})_{ij}| \geq |\lambda|^k \theta^{n-1}.$$

Therefore $\kappa_2(X_n)\rho(A)^k/\|A^k\|_2 \to \infty$ as $\theta \to \infty$. There is still the question of whether X_n is optimally scaled. We can perturb the zero eigenvalues of A to distinct, sufficiently small numbers so that $\|A^k\|_2$ is essentially unchanged and so that the only freedom in X_n is a diagonal scaling. Since X_n has columns of unit 2-norm, $\min_{F=\mathrm{diag}(f_i)} \kappa_2(X_n F) \geq n^{-1/2}\kappa_2(X_n)$ (Theorem 7.5), so even for the optimally scaled X_n the bound can be arbitrarily poor.

18.2. A simple example is

$$A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ \alpha & -\alpha & 1 \end{bmatrix},$$

for which $\lambda(A) = \{-1, 1, 3\}$ and $\lambda(|A|) = \{-1, 2 \pm (4\alpha + 1)^{1/2}\}$, so $\rho(|A|)/\rho(A) \to \infty$ as $\alpha \to \infty$.

19.1. For the Householder matrix $I - (2/v^T v)vv^T$ there is an eigenvalue -1 with eigenvector v, and there are $n - 1$ eigenvalues 1, with eigenvectors any basis for $\mathrm{span}(v)^\perp$. A Givens matrix $G(i, j, \theta)$ has eigenvalues $e^{\pm i\theta}$, together with $n - 2$ eigenvalues 1.

19.2. Straightforward manipulation shows that a bound holds of the form $cnu + O(u^2)$, where c is a constant of order 10.

19.3. We must have $x^*Px = x^*y$, so x^*y must be real. Also, we need $x^*x = y^*y$ and $x \neq y$. Then, with $v = x - y$, $v^*v = 2v^*x$, so $Px = y$.

LAPACK uses modified Householder matrices $P = I - \tau vv^*$ that are unitary but not Hermitian (τ is not real). The benefit of this approach is that such a P can always be chosen so that $Px = \alpha e_1$, with α real (for this equation to hold for a genuine Householder matrix P it would be necessary that x_1 be real). For more details see Lehoucq [780, 1996].

19.4. False. $\det(P) = -1$ for any Householder matrix and $\det(G) = 1$ for any Givens matrix, so $\det(Q) = 1$. Moreover, whereas P is generally a full, symmetric matrix, Q has some zero entries and is generally nonsymmetric.

19.5. The inequalities follow from the fact that, in the notation of (19.3), $\|x_k\|_2 = |r_{kk}|$ and $\|c_k(:, j)\|_2^2 = \sum_{i=k}^{j} r_{ij}^2$, the latter equality being a consequence of the invariance of the 2-norm under orthogonal transformations. The last part follows because QR factorization with column pivoting on A is essentially equivalent to Cholesky factorization with complete pivoting applied to A^TA.

19.6. The second inequality in (19.38) follows from (19.1) and the first from

$$v_k^T v_k = |2\sigma_k(\sigma_k - a_{kk}^{(k)})| \geq 2\sigma_k^2 = 2\|a_k^{(k)}(k\colon m)\|_2^2.$$

From the proof of Lemma 19.2 we see that (19.39) holds with $f_j^{(k)}(1\colon k-1) = 0$ and

$$|f_j^{(k)}| \leq u|\widehat{a}_j^{(k)}| + \tilde{\gamma}_{m-k}\big(|\beta_k||v_k|^T|\widehat{a}_j^{(k)}|\big)|v_k|.$$

Now

$$\begin{aligned}\big(|\beta_k||v_k|^T|\widehat{a}_j^{(k)}|\big)|v_k| &= 2\frac{|v_k|^T|\widehat{a}_j^{(k)}|}{\|v_k\|_2^2}|v_k| \\ &\le 2\frac{\|\widehat{a}_j^{(k)}(k\colon m)\|_2}{\|v_k(k\colon m)\|_2}|v_k| \\ &\le \sqrt{2}\frac{\|\widehat{a}_j^{(k)}(k\colon m)\|_2}{\|\widehat{a}_k^{(k)}(k\colon m)\|_2}|v_k|,\end{aligned}$$

using (19.38), and (19.40) follows. Consider a badly row-scaled problem and $k = 1$, and assume that v_1 has the same scaling as $a_j^{(1)}$ (which row pivoting or row sorting both ensure). Then the bound (19.40) shows that the error will scale in the same way as long as $\|\widehat{a}_j^{(1)}(1\colon m)\|_2/\|\widehat{a}_1^{(1)}(1\colon m)\|_2$ is not too large; column pivoting guarantees that this ratio is bounded by 1. For the full development of this argument see Higham [614, 2000].

19.7. If $y = |W|x$ and W comprises r disjoint rotations then, in a rather loose notation,

$$\begin{aligned}\|y\|_2^2 &\le \sum_{k=1}^{r}\left\|\left|\begin{bmatrix} c_k & s_k \\ -s_k & c_k\end{bmatrix}\right|\begin{bmatrix} x_{i_k} \\ x_{j_k}\end{bmatrix}\right\|_2^2 + \sum_{k=2r+1}^{m} x_{i_k}^2 \\ &\le 2\sum_{k=1}^{2r} x_{i_k}^2 + \sum_{k=2r+1}^{m} x_{i_k}^2 \le 2\|x\|_2^2.\end{aligned}$$

19.8. Straightforward. This problem shows that the CGS and MGS methods correspond to two different ways of representing the orthogonal projection onto $\mathrm{span}\{q_1,\ldots,q_j\}$.

19.9. Assume, without loss of generality, that $\|a_1\|_2 \le \|a_2\|_2$. If E is any matrix such that $A+E$ is rank deficient then

$$0 = \sigma_{\min}(A+E) \ge \sigma_{\min}(A) - \|E\|_2 \quad \Rightarrow \quad \sigma_{\min}(A) \le \|E\|_2.$$

We take $E = [e_1,\, 0]$, where e_1 is chosen so that $e_1^T a_1 = 0$ and $a_1 + e_1 = \alpha a_2$, for some α. From Pythagoras's theorem we have that $\|e_1\|_2 = \tan\theta\,\|a_1\|_2$, and so

$$\sigma_{\min}(A) \le \tan\theta\,\|a_1\|_2 = \tan\theta\,\min(\|a_1\|_2, \|a_2\|_2).$$

Together with the trivial bound $\sigma_{\max}(A) = \|A\|_2 \ge \max(\|a_1\|_2, \|a_2\|_2)$, this yields the result.

19.10. We find that

$$\widehat{Q}_{CGS} = \begin{bmatrix} 1 & 0 & 0 \\ \epsilon & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \\ 0 & \frac{1}{\sqrt{2}} & 0 \\ 0 & 0 & \frac{1}{\sqrt{2}}\end{bmatrix}, \qquad \widehat{Q}_{MGS} = \begin{bmatrix} 1 & 0 & 0 \\ \epsilon & -\frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{6}} \\ 0 & 0 & \sqrt{\frac{2}{3}}\end{bmatrix}.$$

For $\widehat{Q}_{MGS}$, $|\widehat{q}_i^T\widehat{q}_j| \le \epsilon/\sqrt{2}$ for $i \ne j$, but for $\widehat{Q}_{CGS}$, $|\widehat{q}_2^T\widehat{q}_3| = 1/2$. It is easy to see that $fl(1+\epsilon^2) = 1$ implies $\epsilon \le \sqrt{u}$ and that $\kappa_2(A) = \epsilon^{-1}\sqrt{3+\epsilon^2}$. Hence for $\widehat{Q}_{MGS}$, $|\widehat{q}_i^T\widehat{q}_j| \lesssim \kappa_2(A)u/\sqrt{6}$, showing, as expected, that the loss of orthogonality for MGS is bounded in terms of $\kappa_2(A)u$.

19.11. P is the product $P_n \dots P_1$, where P_k is defined in (19.28). Since $v_k^T v_{k-1} = e_k^T e_{k-1} + q_k^T q_{k-1} = 0$, we have

$$P_k P_{k-1} = (I - v_k v_k^T)(I - v_{k-1} v_{k-1}^T) = I - v_k v_k^T - v_{k-1} v_{k-1}^T.$$

Hence

$$P = I - \sum_{k=1}^{n} v_k v_k^T = I - \sum_{k=1}^{n} \begin{bmatrix} e_k e_k^T & -e_k q_k^T \\ -q_k e_k^T & q_k q_k^T \end{bmatrix}.$$

This is of the required form, with $Q = [q_1, \dots, q_n]$ the matrix from the MGS method applied to A.

19.12. With Q defined as in the hint,

$$\begin{aligned} \Delta A = QR - A &= (Q - P_{21})R + \Delta A_2 \\ &= (VW^T - P_{21})R + \Delta A_2 \\ &= V(I - S)W^T R + \Delta A_2 \\ &= V(I + S)^{-1} C^2 W^T R + \Delta A_2 \\ &= V(I + S)^{-1} W^T \cdot WCU^T \cdot UCW^T \cdot R + \Delta A_2 \\ &= V(I + S)^{-1} W^T \cdot P_{11}^T P_{11} R + \Delta A_2 \\ &= V(I + S)^{-1} W^T \cdot P_{11}^T \Delta A_1 + \Delta A_2, \end{aligned}$$

and the result follows.

19.13. To produce the Householder method's P we have to explicitly form the product of the individual Householder transformations. As long as this is done in a stable way the computed $\widehat{P}$ is guaranteed to be nearly orthogonal. MGS's $\widehat{Q}$ is formed in an algebraically different way, and the rounding errors in its formation are different from those in the formation of P; in other words, $\widehat{Q}$ is not a submatrix of $\widehat{P}$. Consequently, there is no reason to expect $\widehat{Q}$ to be nearly orthonormal. Further insight can be gained by a detailed look at the structure of the computed $\widehat{P}$; see Björck and Paige [131, 1992, §4].

19.14. It is straightforward to show that $A^T A - I = (A - U)^T (A + U)$. Taking norms gives the lower bound. Since $A + U = U(H + I)$ we have, from the previous relation,

$$(A - U)^T U = (A^T A - I)(H + I)^{-1}.$$

Hence

$$\|A - U\|_2 = \|(A - U)^T U\|_2 \le \|A^T A - I\|_2 \|(H + I)^{-1}\|_2 \le \frac{\|A^T A - I\|_2}{1 + \sigma_{\min}(A)},$$

since the eigenvalues of H are the singular values of A. In fact, the result holds for any unitarily invariant norm (but the denominators must be left unchanged).

20.1. One approach is to let x be a solution and y an arbitrary vector, and consider $f(\alpha) = \|A(x + \alpha y) - b\|_2^2$. By expanding this expression it can be seen that if the normal equations are not satisfied then α and y can be chosen so that $f(\alpha) < f(0)$. Alternatively, note that for $f(x) = (b - Ax)^T (b - Ax) = x^T A^T Ax - 2b^T Ax + b^T b$ we have $\nabla f(x) = 2(A^T Ax - A^T b)$ and $\nabla^2 f(x) = 2A^T A$, so any solution of the normal equations is a local minimum of f, and hence a global minimum since f is quadratic. The normal equations can be written as $(b - Ax)^T A = 0$, which shows that the residual $b - Ax$ is orthogonal to range(A).

20.2. $A = Q\left[\begin{smallmatrix} R \\ 0 \end{smallmatrix}\right]$. The computed upper triangular QR factor $\widehat{R}$ satisfies $A + \Delta A = Q\left[\begin{smallmatrix} \widehat{R} \\ 0 \end{smallmatrix}\right]$ with $\|\Delta a_j\|_2 \le \tilde{\gamma}_{mn}\|a_j\|_2$, by Theorem 19.4. By Lemma 19.3, the computed transformed right-hand side satisfies $\widehat{c} = Q^T(b + \Delta b)$, with $\|\Delta b\|_2 \le \tilde{\gamma}_{mn}\|b\|_2$.

By Theorem 8.5, the computed solution $\widehat{x}$ to the triangular system $\widehat{R}x = \widehat{c}$ satisfies

$$(\widehat{R} + \Delta R)\widehat{x} = \widehat{c}, \qquad |\Delta R| \le \gamma_n |\widehat{R}|.$$

So $\widehat{x}$ exactly solves the LS problem

$$\begin{bmatrix} \widehat{R} + \Delta R \\ 0 \end{bmatrix} x \approx \begin{bmatrix} \widehat{c} \\ \widehat{d} \end{bmatrix},$$

which is equivalent to the LS problem, on premultiplying by Q,

$$(A + \overline{\Delta A})x := \left(A + \Delta A + Q \begin{bmatrix} \Delta R \\ 0 \end{bmatrix} \right) x \approx b + \Delta b.$$

We have

$$\begin{aligned} \|\overline{\Delta a}_j\|_2 &\le \|\Delta a_j\|_2 + \gamma_n \|\widehat{r}_j\|_2 \\ &= \|\Delta a_j\|_2 + \gamma_n \|a_j + \Delta a_j\|_2 \\ &\le \tilde{\gamma}_{mn} \|a_j\|_2. \qquad \square \end{aligned}$$

20.3. A straightforward verification.

20.4. Notice that

$$\|AX - I_m\|_F^2 = \|A[x_1, \dots, x_m] - [e_1, \dots, e_m]\|_F^2 = \sum_{i=1}^m \|Ax_i - e_i\|_2^2,$$

which is minimized if $\|Ax_i - e_i\|_2$ is minimized for $i = 1\colon m$. Thus we have m independent LS problems. The solution is $x_i = A^+ e_i$, $i = 1\colon m$, that is, $X = A^+ I_m = A^+$. This is the unique solution only if A has full rank, but it is always the unique minimum Frobenius norm solution.

20.5. By Theorem 19.13 there is an orthonormal matrix $[W_1,\ w_{n+1}] \in \mathbb{R}^{m\times(n+1)}$ such that

$$[\, A + \Delta A \quad b + \Delta b \,] = [\, W_1 \quad w_{n+1} \,] \begin{bmatrix} \widehat{R} & \widehat{z} \\ 0 & \widehat{\rho} \end{bmatrix},$$

where $\|\Delta a_j\|_2 \le c_{m,n}u\|a_j\|_2$ and $\|\Delta b\|_2 \le c_{m,n}u\|b\|_2$. The computed solution $\widehat{x}$ to $\widehat{R}x = \widehat{z}$ satisfies

$$(\widehat{R} + \Delta R)\widehat{x} = \widehat{z}, \qquad |\Delta R| \le \gamma_n |\widehat{R}|.$$

Therefore $\widehat{x}$ exactly solves the LS problem

$$\begin{bmatrix} \widehat{R} + \Delta R \\ 0 \end{bmatrix} x \approx \begin{bmatrix} \widehat{z} \\ \widehat{\rho} \end{bmatrix},$$

which, on premultiplying by $[W_1,\ w_{n+1}]$, is equivalent to the LS problem

$$W_1(\widehat{R} + \Delta \widehat{R})x \approx W_1\widehat{z} + w_{n+1}\widehat{\rho},$$

or

$$(A + \overline{\Delta A})x := (A + \Delta A + W_1 \Delta R)x \approx b + \Delta b.$$

We have

$$\|\overline{\Delta a}_j\|_2 \le \|\Delta a_j\|_2 + \gamma_n \|\widehat{r}_j\|_2 \le c_{m,n}u\|a_j\|_2.$$

20.6. Subtracting $\widehat{r}^T\widehat{r} = b^T\widehat{r} - \widehat{x}^T A^T\widehat{r}$ from $r^T r = r^T b$, we have

$$\begin{aligned} r^T r - \widehat{r}^T\widehat{r} &= b^T(r-\widehat{r}) + \widehat{x}^T A^T\widehat{r} \\ &= b^T A(\widehat{x}-x) + \widehat{x}^T(A^T b - A^T A\widehat{x}) \\ &= x^T A^T A(\widehat{x}-x) + \widehat{x}^T(A^T b - A^T A\widehat{x}) \\ &= (x-\widehat{x})^T(A^T A\widehat{x} - A^T b) \\ &= (x-\widehat{x})^T(\Delta c - \Delta A\widehat{x}) \qquad \text{by (20.12).} \end{aligned}$$

Taking norms gives the result.

20.7. By constructing a block LU factorization it is straightforward to show that

$$\det(C(\alpha)-\lambda I) = (\alpha-\lambda)^{m-n}\det(A^T A + \lambda(\alpha-\lambda)I).$$

Hence the eigenvalues of $C(\alpha)$ are $\lambda = \alpha$ ($m-n$ times) together with the solutions of $\lambda(\alpha-\lambda) = -\sigma_i^2$, namely, $\lambda = \alpha/2 \pm (\alpha^2/4+\sigma_i^2)^{1/2}$, $i = 1\colon n$.

Now

$$\sigma_{\min}(C(\alpha)) = \min\Big\{\, \alpha, \Big(\frac{\alpha^2}{4}+\sigma_n^2\Big)^{1/2} - \frac{\alpha}{2} \,\Big\},$$

so the minimum condition number must occur when $\alpha = (\alpha^2/4+\sigma_n^2)^{1/2} - \alpha/2$. This gives $\alpha = \sigma_n/\sqrt{2}$, for which

$$\kappa_2(C(\alpha)) = \frac{1}{2} + \Big(\frac{1}{4} + 2\kappa_2(A)^2\Big)^{1/2}.$$

The lower bound for the maximum is achieved for $\alpha = \sigma_1/\sqrt{2}$.

20.8. Let $y = 0$ and $\|(A+\Delta A)y - b\|_2 = \min$. If $b = 0$ then we can take $\Delta A = 0$. Otherwise, the normal equations tell us that $(A+\Delta A)^T b = 0$, so $\|\Delta A\|_2 \ge \|A^T b\|_2/\|b\|_2$. This lower bound is achieved, and the normal equations satisfied, for $\Delta A^T = -(A^T b)b^T/b^T b$. Hence, for $\theta = \infty$,

$$\eta_F(0) = \begin{cases} \|A^T b\|_2/\|b\|_2, & b \ne 0, \\ 0, & b = 0. \end{cases}$$

20.9. For the case $\lambda_* < 0$ we have

$$\begin{aligned} \eta_F(y) &= \lambda_{\min}\left(AA^T - \mu\frac{rr^T}{\|y\|_2^2} + \mu\frac{\|r\|_2^2}{\|y\|_2^2}I\right)^{1/2} \\ &= \lambda_{\min}\left(AA^T + \mu\frac{\|r\|_2^2}{\|y\|_2^2}\Big(I - \frac{rr^T}{\|r\|_2^2}\Big)\right)^{1/2} \\ &= \lambda_{\min}\left(AA^T + \mu\frac{\|r\|_2^2}{\|y\|_2^2}\Big(I - \frac{rr^T}{\|r\|_2^2}\Big)^2\right)^{1/2} \\ &= \sigma_{\min}\left(\Big[A \quad \sqrt{\mu}\frac{\|r\|_2}{\|y\|_2}\Big(I - \frac{rr^T}{\|r\|_2^2}\Big)\Big]\right). \end{aligned}$$

20.10. This is a standard result that can be derived by considering the first- and second-order optimality conditions in the theory of constrained optimization.

20.11. Setting $B = 0$ and $d = 0$ and using $\|b\|_2 \le \|r\|_2 + \|A\|_F\|x\|_2$ we obtain

$$\frac{\|\Delta x\|_2}{\|x\|_2} \le \epsilon\widetilde{\kappa}(A)\Big(2 + (\widetilde{\kappa}(A)+1)\frac{\|r\|_2}{\|A\|_F\|x\|_2}\Big) + O(\epsilon^2),$$

where $\widetilde{\kappa}(A) = \|A\|_F\|A^+\|_2$, which is a first-order approximation of (20.1).

21.1. Setting the gradient of (21.13) to zero gives $A^TAx - A^Tb + c = 0$, which can be written as $y = b - Ax$, $A^Ty = c$, which is (21.12). The Lagrangian for (21.14) is $L(y,x) = \frac{1}{2}(y-b)^T(y-b) + x^T(A^Ty - c)$. We have $\nabla_y L(y,x) = y - b + Ax$ and $\nabla_x L(y,x) = A^Ty - c$. Setting the gradients to zero gives the system (21.12).

22.1. The modified version has the same flop count as the original version.

22.4. The summation gives

$$\sum_{j=0}^{n} x^j \sum_{i=0}^{n} w_{ij} = \sum_{i=0}^{n} l_i(x) \equiv 1,$$

which implies the desired equality. It follows that all columns of V^{-1} except the first must have both positive and negative entries. In particular, $V^{-1} \geq 0$ is not possible. The elements of V^{-1} sum to 1, independently of the points α_i (see also Problem 14.7).

22.5. We have $U(i,:)T(\alpha_0,\ldots,\alpha_n) = W(i,:)V(\alpha_0,\ldots,\alpha_n)$. But $T = LV$, where L has the form illustrated for $n = 5$ by

$$L = \begin{bmatrix} 1 & & & & \\ 0 & 1 & & & \\ -1 & 0 & 2 & & \\ 0 & -3 & 0 & 4 & \\ 1 & 0 & -8 & 0 & 8 \end{bmatrix}$$

and $Le = e$, so $U(i,:)L = W(i,:)$, or $U(i,:)^T = L^{-T}W(i,:)^T$. But $L^{-T} \geq 0$ by the given x^n formula, so $\|L^{-T}\|_1 = \|L^{-1}\|_\infty = \|L^{-1}e\|_\infty = \|e\|_\infty = 1$, hence $\|U(i,:)\|_1 \leq \|W(i,:)\|_1$.

As an aside, we can evaluate $\|L\|_\infty$ as

$$\|L\|_\infty = |T_n(\sqrt{-1})| = \frac{1}{2}\big[(1+\sqrt{2})^n + (1-\sqrt{2})^n\big],$$

after a little trigonometric algebra.

22.7. Denote the matrix by C. For the zeros of T_{n+1} we have

$$CC^T = (n+1)\,\mathrm{diag}\left(1, \frac{1}{2}, \frac{1}{2}, \ldots, \frac{1}{2}\right).$$

It follows that $\|C\|_2^2 = \sqrt{n+1}$, $\|C^{-1}\|_2^2 = 2/\sqrt{n+1}$, giving the result.

Extrema of T_n: we have

$$CDC^T = \frac{n}{2}D^{-1}, \qquad D = \mathrm{diag}\left(\frac{1}{2}, 1, 1, \ldots, 1, \frac{1}{2}\right).$$

Hence $B = CD^{1/2}$ satisfies $BB^T = \frac{n}{2}D^{-1}$, and so $\kappa_2(B) = \kappa_2(D)^{1/2} = \sqrt{2}$. Then $\kappa_2(C) \leq \kappa_2(D^{1/2})\kappa_2(B) = 2$.

22.8. The first expression follows from the formula (15.10) for the inverse of an upper bidiagonal matrix. The expression shows that a small componentwise relative change in a bidiagonal matrix causes only a small componentwise relative change in the inverse. For the second part, we note that in Algorithm 22.2, for the monomials, we have $U_k + \Delta U_k = \mathrm{diag}(1+\epsilon_i)\widetilde{U}_k$, $|\epsilon_i| \leq u$, where $\widetilde{U}_k$ agrees with U_k everywhere except in certain superdiagonal elements, where $(\widetilde{U}_k)_{i,i+1} = (U_k)_{i,i+1}(1+\delta_{i,i+1})$, $|\delta_{i,i+1}| \leq u$. The result now follows by applying the first part (and noting that U_k is of dimension $n+1$).

22.9. The increasing ordering is never produced, since the algorithm must choose α_1 to maximize $|\alpha_1 - \alpha_0|$.

22.11. The dual condition number is

$$\frac{\| \, |V^{-T}||V^T \operatorname{diag}(0,1,2,\ldots,n)a| \, \|_\infty}{\|a\|_\infty}.$$

See Higham [581, 1987] for proofs.

23.1. Consider the case where $m \le \min(n,p)$, and suppose $n = jm$ and $p = km$ for some integers j and k. Then the multiplication of the two matrices can be split into $m \times m$ blocks:

$$AB = [\, A_1 \quad A_2 \quad \ldots \quad A_j \,] \begin{bmatrix} B_{11} & \ldots & B_{1k} \\ \vdots & & \vdots \\ B_{j1} & \ldots & B_{jk} \end{bmatrix},$$

which involves a total of jk multiplications of $m \times m$ matrices, each involving $O(m^\alpha)$ operations. Thus the total number of operations is $O(jkm^\alpha)$, or $O(m^{\alpha-2}np)$, as required, and we can show similar results for the cases when n and p are the smallest dimensions.

23.2. $\frac{1}{2}n^3 + n^2$ multiplications and $\frac{3}{2}n^3 + 2n(n-1)$ additions.

23.3. For large $n = 2^k$, $S_n(8)/S_n(n) \approx 1.96 \times (7/8)^k$ and $S_n(1)/S_n(8) \approx 1.79$. The ratio $S_n(8)/S_n(n)$ measures the best possible reduction in the amount of arithmetic by using Strassen's method in place of conventional multiplication. The ratio $S_n(1)/S_n(8)$ measures how much more arithmetic is used by recurring down to the scalar level instead of stopping once the optimal dimension n_0 is reached. Of course, the efficiency of a practical code for Strassen's method also depends on the various non-floating point operations.

23.5. Apart from the differences in stability, the key difference is that Winograd's formula relies on commutativity and so cannot be generalized to matrices.

23.7. Some brief comments are given by Douglas, Heroux, Slishman, and Smith [351, 1994].

23.9. The inverse is

$$\begin{bmatrix} I & -A & AB \\ 0 & I & -B \\ 0 & 0 & I \end{bmatrix}.$$

Hence we can form AB by inverting a matrix of thrice the dimension. This result is not of practical value, but it is useful for computational complexity analysis.

25.1. (a) We have

$$\|x_{k+1} - a\| = \|G(x_k) - a + e_k\| \le \beta\|x_k - a\| + \alpha. \tag{A.15}$$

If $\|x_k - a\| \le \alpha/(1-\beta)$ then

$$\|x_{k+1} - a\| \le \beta\frac{\alpha}{1-\beta} + \alpha = \frac{\alpha}{1-\beta}.$$

If $\|x_k - a\| > \alpha/(1-\beta)$, so that $\beta\|x_k - a\| < \|x_k - a\| - \alpha$, then, using (A.15),

$$\|x_{k+1} - a\| < (\|x_k - a\| - \alpha) + \alpha = \|x_k - a\|.$$

(b) We break the proof into two cases. First, suppose that $\|x_m - a\| \leq \alpha/(1-\beta)$ for some m. By part (a), the same inequality holds for all $k \geq m$ and we are finished. Otherwise, $\|x_k - a\| > \alpha/(1-\beta)$ for all k. By part (a) the positive sequence $\|x_k - a\|$ is monotone decreasing and therefore it converges to a limit l. Suppose that $l = \alpha/(1-\beta)+\delta$, where $\delta > 0$. Since $\beta < 1$, for some k we must have

$$\|x_k - a\| < \frac{\alpha}{1-\beta} + \frac{\delta}{\beta}.$$

By (A.15),

$$\|x_{k+1} - a\| < \frac{\beta\alpha}{1-\beta} + \delta + \alpha = \frac{\alpha}{1-\beta} + \delta = l,$$

which is a contradiction. Hence $l = \alpha/(1-\beta)$, as required.

(c) The vectors e_k can be regarded as representing the rounding errors on the kth iteration. The bound in (b) tells us that provided the error vectors are bounded by α, we can expect to achieve a relative error no larger than $\alpha/(1-\beta)$. In other words, the result gives us an upper bound on the achievable accuracy. When this result is applied to stationary iteration, β is a norm of the iteration matrix; to make β less than 1 we may have to use a scaled norm of the form $\|A\| := \|XAX^{-1}\|$.

26.2. With $n = 3$ and almost any starting data, the backward error can easily be made of order 1, showing that the method is unstable. However, the backward error is found to be roughly of order $\kappa_\infty(A)u$, so the method may have a backward error bound proportional to $\kappa_\infty(A)$ (this is an open question).

27.1. (b) An optimizing compiler might convert the test `xp1 > 1` to `x+1 > 1` and then to `x > 0`. (For a way to stop this conversion in Fortran, see the solution to Problem 27.3.) Then the code would compute a number of order $2^{e_{\min}}$ instead of a number of order 2^{-t}.

27.3. The algorithm is based on the fact that the positive integers that can be exactly represented are $1, 2, \ldots, \beta^t$ and

$$\beta^t + \beta, \beta^t + 2\beta, \beta^t + 3\beta, \ldots, \beta^{t+1}, \beta^{t+1} + \beta^2, \ldots.$$

In the interval $[\beta^t, \beta^{t+1}]$ the floating point numbers are spaced β apart. This interval must contain a power of 2, $a = 2^k$. The first while loop finds such an a (or, rather, the floating point representation of such an a) by testing successive powers 2^i to see if both 2^i and 2^i+1 are representable. The next while loop adds successive powers of 2 until the next floating point number is found; on subtracting a the base β is produced. Finally t is determined as the smallest power of β for which the distance to the next floating point number exceeds 1.

The routine can fail for at least two reasons. First, an optimizing compiler might simplify the test `while (a+1) - a == 1` to `while 1 == 1`, thus altering the meaning of the program. Second, in the same test the result `(a+1)` might be held in an extra length register and the subtraction done in extra precision. The computation would then reflect this higher precision and not the intended one. We could try to overcome this problem by saving the result `a+1` in a variable, but an optimizing compiler might undo this effort by dispensing with the variable and storing the result in a register. In Fortran, the compiler's unwanted efforts can be nullified by a test of the form `while foo(a+1) - a == 1`, where `foo` is a function that simply returns its argument. The problems caused by the use of extra length registers were discussed by Gentleman and Marovich [476, 1974]; see also Miller [853, 1984, §2.2].

27.4. (This description is based on Schreiber [1022, 1989].) The random number generator in `matgen` repeats after 16384 numbers [744, 1998, §3.2.1.2, Thm. B]. The dimension $n = 512$ divides the period of the generator ($16384 = 512 \times 32$), with the effect that the first 32 columns of the matrix are repeated 16 times ($512 = 32 \times 16$), so the matrix has this structure:

```
B = rand(512,32);
A = [B, B, B, B, B, B, B, B, B, B, B, B, B, B, B, B];
```

Now consider the first 32 steps of Gaussian elimination. We apply 32 transformations to A that have the effect, in exact arithmetic, of making B upper trapezoidal. In floating point arithmetic, they leave a residue of small numbers (about $u \approx 10^{-7}$ in size) in rows 33 onward. Because of the structure of the matrix, identical small numbers occur in each of the 15 blocks of A to the right of the first. Thus, the remaining $(512-32) \times (512-32)$ submatrix has the same block structure as A (but with 15 block columns). Hence this process repeats every 32 steps:

after 32 steps the elements drop to $\approx 10^{-7}$;
after 64 steps the elements drop to $\approx 10^{-14}$;
after 96 steps the elements drop to $\approx 10^{-21}$;
after 128 steps the elements drop to $\approx 10^{-28}$;
after 160 steps the elements drop to $\approx 10^{-35}$;
after 192 steps the elements would drop to $\approx 10^{-42}$,

but that is less than the underflow threshold. The actual pivots do not exactly match the analysis, which is probably due to rank deficiency of one of the submatrices generated. Also, underflow occurs earlier than predicted, apparently because two small numbers (both $O(10^{-21})$) are multiplied in a saxpy operation.

27.5. Let s_i and t_i denote the values of s and t at the start of the ith stage of the algorithm. Then

$$\sum_{k=1}^{i-1} x_k^2 = t_i^2 s_i. \tag{A.16}$$

If $|x_i| > t_i$ then the algorithm uses the relation

$$\sum_{k=1}^{i} x_k^2 = t_i^2 s_i + x_i^2 = x_i^2\big((t/x_i)^2 s_i + 1\big),$$

so with $s_{i+1} = (t_i/x_i)^2 s_i + 1$ and $t_{i+1} = |x_i|$, (A.16) continues to hold for the next value of i. The same is true trivially in the case $|x_i| \le t_i$.

This is a one-pass algorithm using n divisions that avoids overflow except, possibly, on the final stage in forming $t\sqrt{s}$, which can overflow only if $\|x\|_2$ does.

27.7. (a) The matrix $A := I + Y^2 = I - Y^TY$ is symmetric positive semidefinite since $\|Y\|_2 \le 1$, hence it has a (unique) symmetric positive semidefinite square root X satisfying $X^2 = A = I + Y^2$. The square root X is a polynomial in A (see, for example, Higham [580, 1987]) and hence a polynomial in Y; therefore X and Y commute, which implies that $(X+Y)^T(X+Y) = (X-Y)(X+Y) = X^2 - Y^2 = I$, as required.

27.8. $|x|/3$ could underflow to zero, or become unnormalized, when $\sqrt{|x|}/\sqrt{3}$ does not.